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PRIMES, FOLDS, AND THE ONE DOT: A COMPUTATIONAL ATLAS OF WEIL POSITIVITY, PRIME-SIEVE GEOMETRY AND THE ZETA ZEROS

CHRISTOPHE MICHAELS

ABSTRACT. We present a computational and visual study of several objects that surround the Riemann Hypothesis (RH), organized around the odd sector of Weil’s quadratic form. Using exact rational kernels and 420 to 640-bit arithmetic we compute the lowest eigenvalue $\lambda(a)$ of the odd Weil form on $[-a, a]$ for $0.1 \leq a \leq 1.1$, and find that each prime power $n \leq 9$ is load-bearing: deleting the most recent one makes the form indefinite within 0.025 of its entry point $a = \frac{1}{2} \log n$ (within 0.003 for $n \geq 5$). For the minimizing test function we verify the explicit formula against 22,491 zeta zeros to a relative accuracy of 2×10^{-5} , show that its Fourier transform places zeros on the zeta zeros below the height $T^*(a) = 2\pi e^{2a}$ (the first to within 3×10^{-32} at $a = 1.1$), and find that zeros above $T^*(a)$ account for all but 10^{-5} of $\lambda(a)$, while $-\log \lambda(a)$ grows by about 1.83 per unit of $T^*(a)$. On the arithmetic side we give closed forms for the upper edge of the Matiyasevich–Stechkin parabola sieve, tabulate every prime gap up to 10^{11} (confirming the Hardy–Littlewood singular-series weights to a median deviation of 2.2%), and measure the logarithmic mean square of $(\psi(x) - x)/\sqrt{x}$ up to 10^{10} , obtaining 0.0458 against Cramér’s value $2 + \gamma - \log 4\pi = 0.04619$ under RH. We also check a set of derivative-jet identities for ξ , contrast the functional-equation symmetry with the Davenport–Heilbronn function, and record several equivalent reformulations of RH in pictorial form. Nothing in this paper proves RH. Every statement is labelled as a theorem from the literature (with citation), an elementary result proved here, a computation (with its precision and cross-checks), or a heuristic.

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1. INTRODUCTION

1.1. Weil positivity. Let $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ be the completed zeta function, and write $F(z) = \xi(\frac{1}{2} + z)$. The nontrivial zeros of ζ are the zeros $\rho = \frac{1}{2} + i\gamma$ of ξ , with $\gamma \in \mathbb{C}$; the Riemann Hypothesis asserts that every γ is real. For an even test function g with Fourier transform $\hat{g}(t) = \int g(x)e^{itx} dx$, Weil's explicit formula [43] reads

$$(1) \quad \sum_{\rho} \hat{g}(\gamma_{\rho}) = \hat{g}\left(\frac{i}{2}\right) + \hat{g}\left(-\frac{i}{2}\right) - \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} (g(\log n) + g(-\log n)) + W_{\infty}(g),$$

where Λ is the von Mangoldt function and W_{∞} is the archimedean term coming from $\Gamma(s/2)$. Taking $g = f * \tilde{f}$ with $\tilde{f}(x) = f(-x)$ and f real and supported in $[-a, a]$ turns the right-hand side into a quadratic form $Q_a(f)$. If RH holds then $\hat{g}(\gamma) = |\hat{f}(\gamma)|^2 \geq 0$ for every zero and $Q_a \geq 0$; Weil's criterion [43, 6] states the converse, so that RH is equivalent to $Q_a \geq 0$ for every $a > 0$.

The kernel of Q_a is even, so Q_a splits orthogonally into an even and an odd sector. Following Zhu [46] we work in the odd sector and write

$$\lambda(a) = \min\{Q_a(f) : f \text{ real, odd, } \text{supp } f \subseteq [-a, a], \|f\|_2 = 1\}.$$

Positivity of $\lambda(a)$ for every a is the form of Weil's criterion used in [46] and in the author's atlas [28], where the odd-sector reduction is attributed to Yoshida [45]. Zhu certifies two-sided bounds on $\lambda(a)$ for $a \leq 0.8$ and identifies a double-exponential decay law related to the Landau–Widom rate for time-band limiting [46]. The present paper computes $\lambda(a)$ and its minimizer to much higher precision than is needed for positivity, and uses the results to draw a number of pictures of the objects around RH.

A second motivation is recent: Alpöge and Furman [1] proved unconditionally that more than two thirds of the zeros are simple and on the critical line, using a finite compression of Weil's Hermitian form together with an inertia count. Section 11 explains how their argument sits inside the same machinery.

1.2. Contents. The sections are largely independent.

- Section 2: high-precision odd Weil floors, and the observation that each prime power $2, 3, 4, 5, 7, 8, 9$ is load-bearing at its entry point (Computation 2.1).
- Section 3: simultaneous crests of the prime comb and their Dirichlet exponents.
- Section 4: the explicit formula drawn on the prime line, including an exact envelope for the prime-power weights (Proposition 4.1).
- Section 5: the parabola sieve; closed forms for its upper edge (Proposition 5.3) and for the “reach-back” chords of each prime (Proposition 5.6).
- Section 6: all prime gaps to 10^{11} drawn as a chain of triangles, a residue alternation (Proposition 6.2), and the Hardy–Littlewood weights.

- Sections 7–9: derivative jets of ξ , the functional-equation fold, the Davenport–Heilbronn control, and several ways of holding every zero at once.
- Sections 10–11: the counting function drawn as a stack of rings, the effect of a hypothetical off-line zero, rigidity of the zeros, and the proportion results.
- Section 12: a “light-cone” reading of the odd Weil form and a numerical test of it (Computation 12.5).
- Section 13: the scaling structure of the prime waves and a measurement of Cramér’s mean-square constant to 10^{10} (Computation 13.3).
- Section 14: open questions. Appendix A describes the engines and their validation; Appendix B lists every statement with its status.

1.3. Status conventions. Results are stated in four kinds of environment. A *Theorem* is a result from the literature and carries a citation. A *Proposition* is an elementary statement proved in the text. A *Computation* is a numerical result, stated with its working precision and the cross-checks it passed; computations beyond the ranges certified in [46] are numerics, not certificates. A *Heuristic* or *Remark* records an interpretation or analogy and makes no mathematical claim. RH remains open throughout.

1.4. Notation. $\Lambda(n)$ is the von Mangoldt function, $\psi(x) = \sum_{n \leq x} \Lambda(n)$, p, q, r denote primes, and $g = q - p$ denotes the gap between consecutive primes $p < q$. For a zero $\rho = \beta + i\gamma$ we call β its real part and γ its height. $N(T)$ is the number of zeros with $0 < \gamma \leq T$. When γ appears without a subscript in $2 + \gamma - \log 4\pi$ it denotes Euler’s constant. Figures use a fixed color convention: 2 blue, 3 orange, 5 green, 7 yellow, magenta for the tooth walk of Section 6, and teal for regions where $|\xi|$ decreases away from the critical line.

2. THE ODD WEIL FORM AND THE PRIME RELAY

2.1. Discretization. In the odd sector we expand $f(x) = a^{-1/2} \sum_{i < K} c_i N_i P_{2i+1}(x/a)$ in normalized odd Legendre polynomials ($N_i = \sqrt{(4i+3)/2}$), so that $\|f\|_2^2 = \sum c_i^2$. The autocorrelation $g = f * \tilde{f}$ is then a quadratic form in c whose kernel involves the exact polynomials $\int_{-1}^{1-s} P_j(u) P_k(u+s) du$, computed in rational arithmetic. The archimedean part is evaluated through the symbol

$$(2) \quad \Psi_a(t) = \operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \log \pi - \sum_{\log n < 2a} \frac{2\Lambda(n)}{\sqrt{n}} \cos(t \log n)$$

(here ψ is the digamma function) in the form given by Zhu [46, Lemma 2.5]. For odd real f the two polar terms in (1) combine to $-2\left(\int f(x) \sinh(x/2) dx\right)^2$. We use $K = 40$ modes at 420-bit precision for $a \leq 0.75$ and $K = 72$ modes at 640 bits for $0.76 \leq a \leq 1.1$; the lowest eigenvalue is found by inverse iteration and the number of negative eigenvalues by an LDL^T inertia count (Sylvester’s law). The resulting floors agree with Zhu’s reference values: $\lambda(0.8) = 1.5599 \times 10^{-14}$ against 1.57×10^{-14} , and $\lambda(1.0) = 1.4913 \times 10^{-26}$ against 1.49×10^{-26} .

2.2. The relay. A prime power n enters Q_a when $2a$ passes $\log n$. To see how much each one matters, we recompute the form with the newest prime power deleted and locate the first a at which the deleted form acquires a negative eigenvalue.

Computation 2.1 (The relay). For each prime power $n \in \{2, 3, 4, 5, 7, 8, 9\}$, the odd form with n deleted turns indefinite shortly after n enters (Table 1, Figures 1 and 2). The full form stays positive definite throughout $0.1 \leq a \leq 1.1$.

Computation 2.2 (Color budget). Group the terms of all powers of p into one “color”. At the minimizer, color 2 (the terms of 2, 4, 8) pays 98.8% of the archimedean-plus-polar deficit at $a = 0.8$

TABLE 1. Entry points and failure points of the deleted form.

Prime power n	Entry $\frac{1}{2} \log n$	Deleted form fails at	Gap	Precision
2	0.346574	0.371601	0.0250	420 bits, 40 modes
3	0.549306	0.557323	0.0080	420 bits, 40 modes
$4 = 2^2$	0.693147	0.699032	0.0059	420 bits, 40 modes
5	0.804719	0.807245	0.0025	640 bits, 72 modes
7	0.972955	0.974246	0.0013	640 bits, 72 modes
$8 = 2^3$	1.039721	1.041389	0.0017	640 bits, 72 modes
$9 = 3^2$	1.098612	1.099731	0.0011	640 bits, 72 modes

and 97.6% at $a = 1.0$; color 3 pays nearly all the rest (2.4% at $a = 1.0$); colors 5 and 7 contribute fractions 3×10^{-8} and 9×10^{-20} at $a = 1.0$.

Remark 2.3. Computation 2.1 is a finite observation: at each tested support the newest prime power is load-bearing. Nothing here shows that the hand-off continues for every a , and $\lambda(a)$ itself decays double-exponentially (Section 12). A proof of positivity would need a mechanism that is uniform in a .

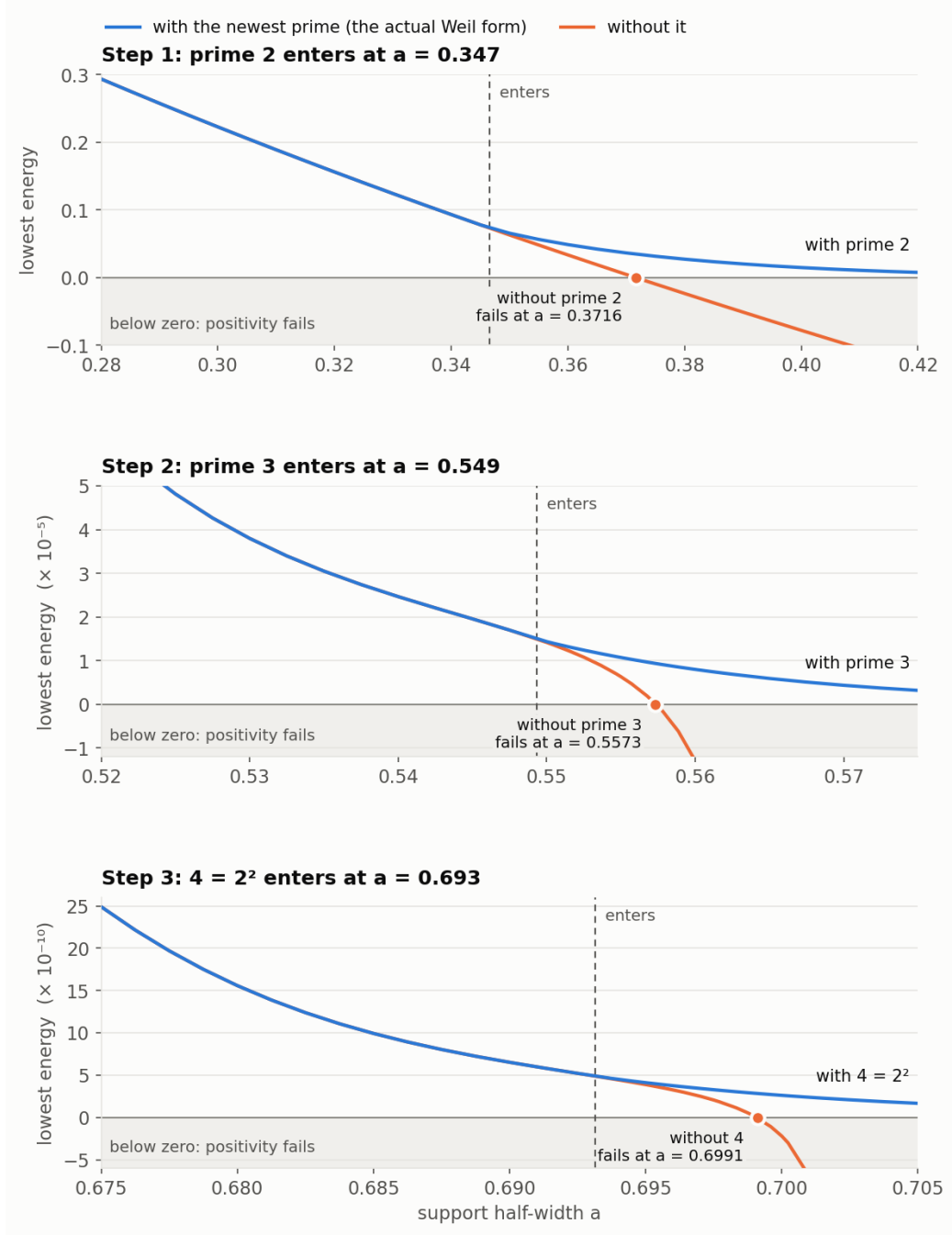


FIGURE 1. The first three hand-offs. Lowest eigenvalue of the odd Weil form with the newest prime power (blue) and with it deleted (orange). Without 2 the floor fails at $a = 0.3716$, without 3 at 0.5573 , and without $4 = 2^2$ at 0.6991 . Double-precision preview with 24 modes; the entries of Table 1 are the high-precision values.

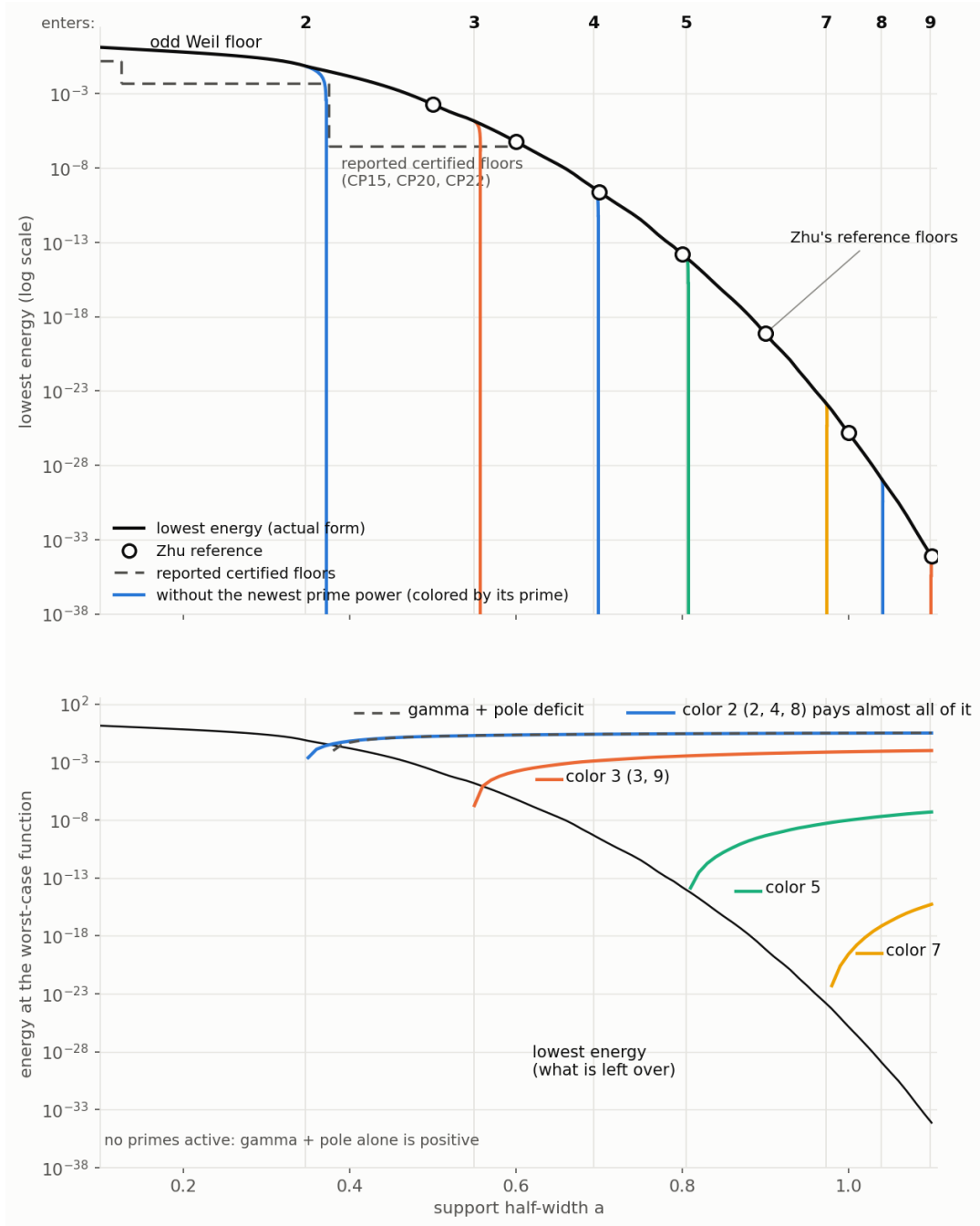


FIGURE 2. Seven prime powers out to $a = 1.1$. Top: $\lambda(a)$ on a logarithmic scale, with Zhu's reference floors [46] and the certified floors reported in [28] (CP15, CP20, CP22); each colored drop is the deleted form. Bottom: at the minimizer, the archimedean and polar terms run a deficit, which the prime-power terms, grouped by prime ("colors"), pay.

3. SIMULTANEOUS CRESTS OF THE PRIME COMB

The prime sum in (2) is a superposition of cosines $\cos(t \log p^k)$, one family (“color”) per prime. At $t = 2\pi k / \log 2$ every term of color 2 is at a crest; in musical language, k is a number of equal steps per octave. The other colors crest there only approximately, and the prime comb is large where they nearly crest together.

Proposition 3.1. *The numbers $\log p$, p prime, are linearly independent over $\mathbb{Q}$. Consequently no $t \neq 0$ is a simultaneous exact crest of two or more colors.*

Proof. A relation $\sum m_p \log p = 0$ with integers m_p gives $\prod p^{m_p} = 1$, which forces all $m_p = 0$ by unique factorization. If $t \log p \in 2\pi\mathbb{Z}$ and $t \log q \in 2\pi\mathbb{Z}$ with $t \neq 0$ then $\log p / \log q \in \mathbb{Q}$, a contradiction. $\square$

By Dirichlet’s simultaneous approximation theorem the worst of the d phase misses at the best k up to K is at most of order $K^{-1/d}$.

Computation 3.2 (Record alignments). Table 2 lists the record tunings, the number of records below 2×10^8 steps, and the fitted decay exponent of the worst miss. For the octave and 3 alone, 399 record tunings were computed out to 209-digit step counts; each record multiplies k by about 3.3 on average, close to Lévy’s constant 3.28 for a typical real number. Replacing $\log 3, \log 5, \log 7$ by random numbers gives the same slopes, so the descent is generic (Figures 3 and 4).

TABLE 2. Record simultaneous alignments with the octave.

Colors with 2	Record step counts k	Records to 2×10^8	Fitted slope	Dirichlet
3	5, 12, 41, 53, 306, 665, 15601, 31867, ...	17	−0.98	−1
3, 5	10, 12, 19, 34, 53, 118, 612, 1783, ...	20	−0.53	−1/2
3, 5, 7	9, 10, 31, 99, 140, 171, 441, 3125, ...	20	−0.33	−1/3
3, 5, 7, 11	9, 15, 41, 46, 72, 224, 270, 1578, ...	23	−0.28	−1/4

Remark 3.3. The dips of the Weil symbol (2) occur at near-alignments of the colors. Proposition 3.1 says these alignments are never exact; the rate at which they improve is the generic Dirichlet rate.

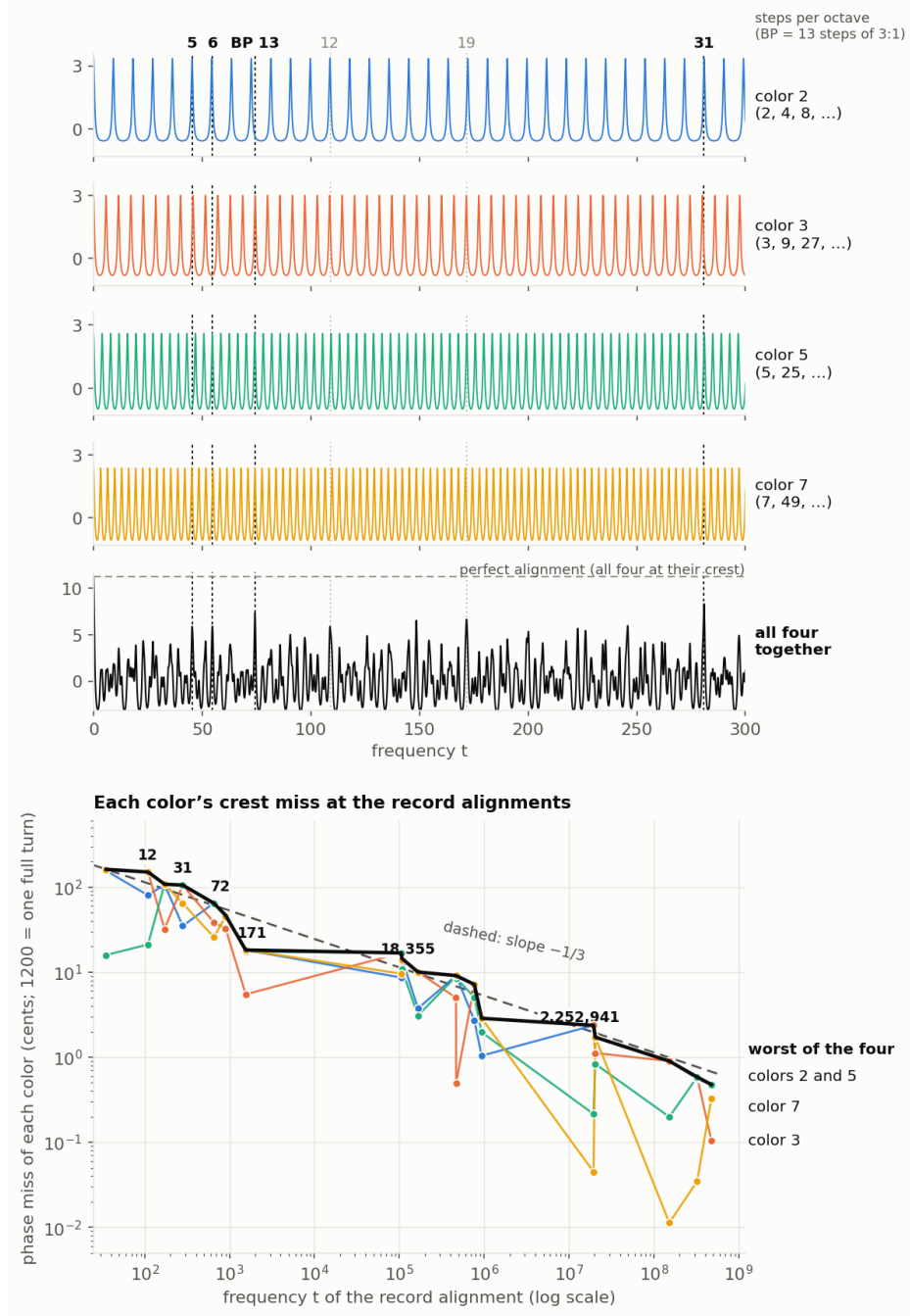


FIGURE 3. Top: the wave of each color $p \in \{2, 3, 5, 7\}$ in the prime comb against t , with record simultaneous alignments marked (5, 6, the 13 steps of 3:1 near $t = 74.26$, and 31 steps per octave). Bottom: at every record alignment up to $t \approx 5 \times 10^8$, the phase miss of each color; the worst of the four follows slope $-1/3$.

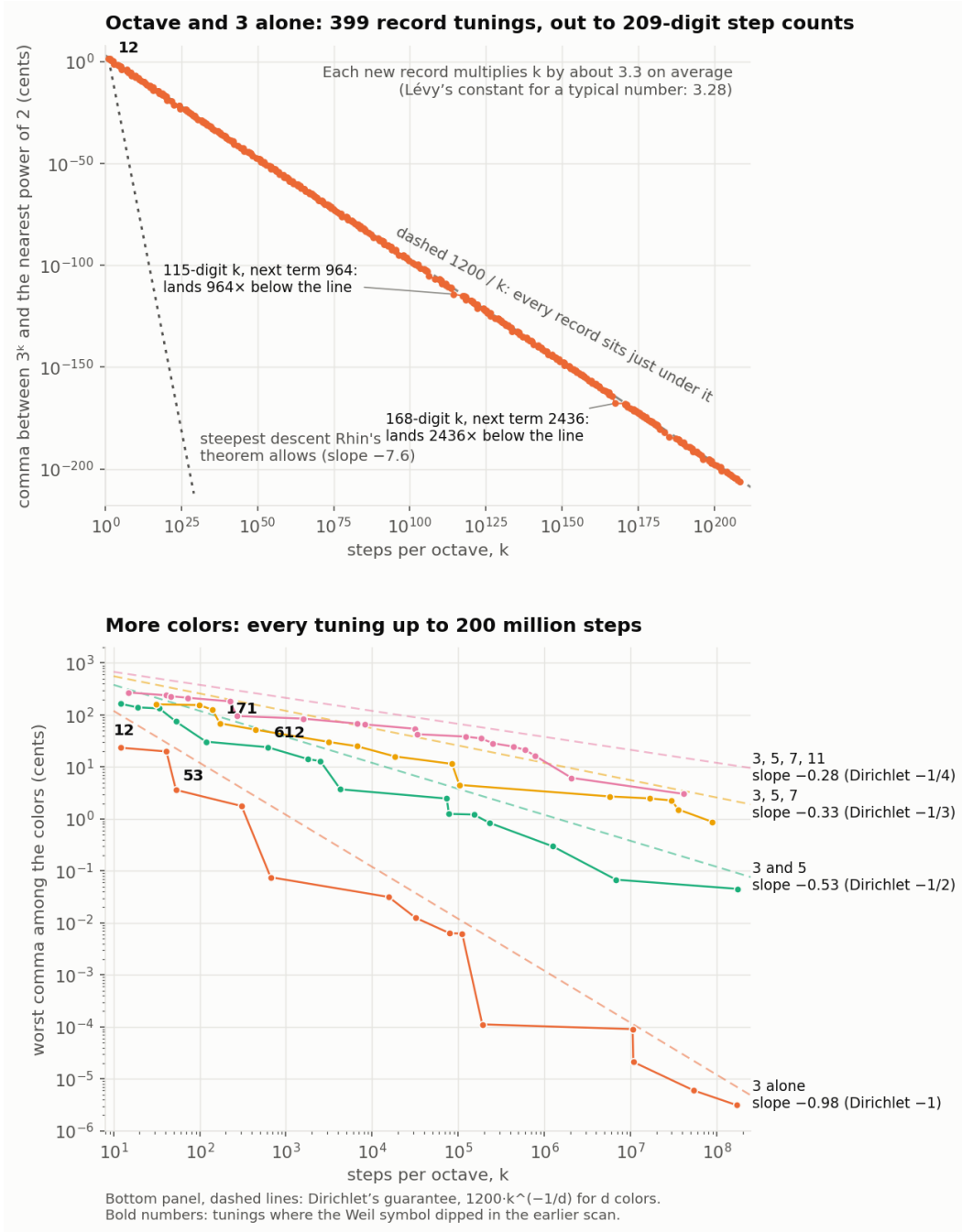


FIGURE 4. Top: the octave and 3 alone, 399 record tunings to 209-digit step counts; every record sits just under $1200/k$ cents, far from the steepest descent permitted by known irrationality measures of $\log 3/\log 2$. Bottom: more colors, every tuning up to 2×10^8 steps, with the Dirichlet exponents $-1/d$ dashed.

4. THE EXPLICIT FORMULA ON THE PRIME LINE

Place each prime power n at position $x = \log n$ with weight $2\Lambda(n)/\sqrt{n}$; this is the prime side of (1) with g a narrow bump. The archimedean side enters through Chebyshev's identity $\log n! = \sum_{d \leq n} \Lambda(d) \lfloor n/d \rfloor$.

Proposition 4.1 (Envelope). *For every prime power p^k , with $x = \log p^k$,*

$$\frac{2\Lambda(p^k)}{\sqrt{p^k}} = \frac{1}{k} \cdot 2x e^{-x/2}.$$

Thus primes lie on the curve $y = 2xe^{-x/2}$, squares on half of it, cubes on a third, and k -th powers on $1/k$ of it. The curve peaks at $x = 2$ ($p = e^2 \approx 7.39$) with height $4/e$.

Proof. $\Lambda(p^k) = \log p = x/k$ and $\sqrt{p^k} = e^{x/2}$. □

Computation 4.2. With every prime power up to 10^6 (78,734 terms), the archimedean energy minus all prime-power waves agrees with a sum of Gaussian bumps centered at the zeta zeros to 5×10^{-14} (Figure 5). Conversely, the smooth term $2e^{x/2}$ minus one cosine wave per zero, using the first 1,000 zeros only, reproduces the height of every prime-power bar up to $n \approx 270$ to 0.1% and vanishes between bars (Figure 7); in the smooth-weighted norm the agreement is 1.6×10^{-6} .

Remark 4.3. The zero side of (1) sees a prime power only through its position $\log n$ and its weight $\Lambda(n)/\sqrt{n}$. In this sense the explicit formula is indifferent to the digits of n and sensitive only to its place on the logarithmic line.

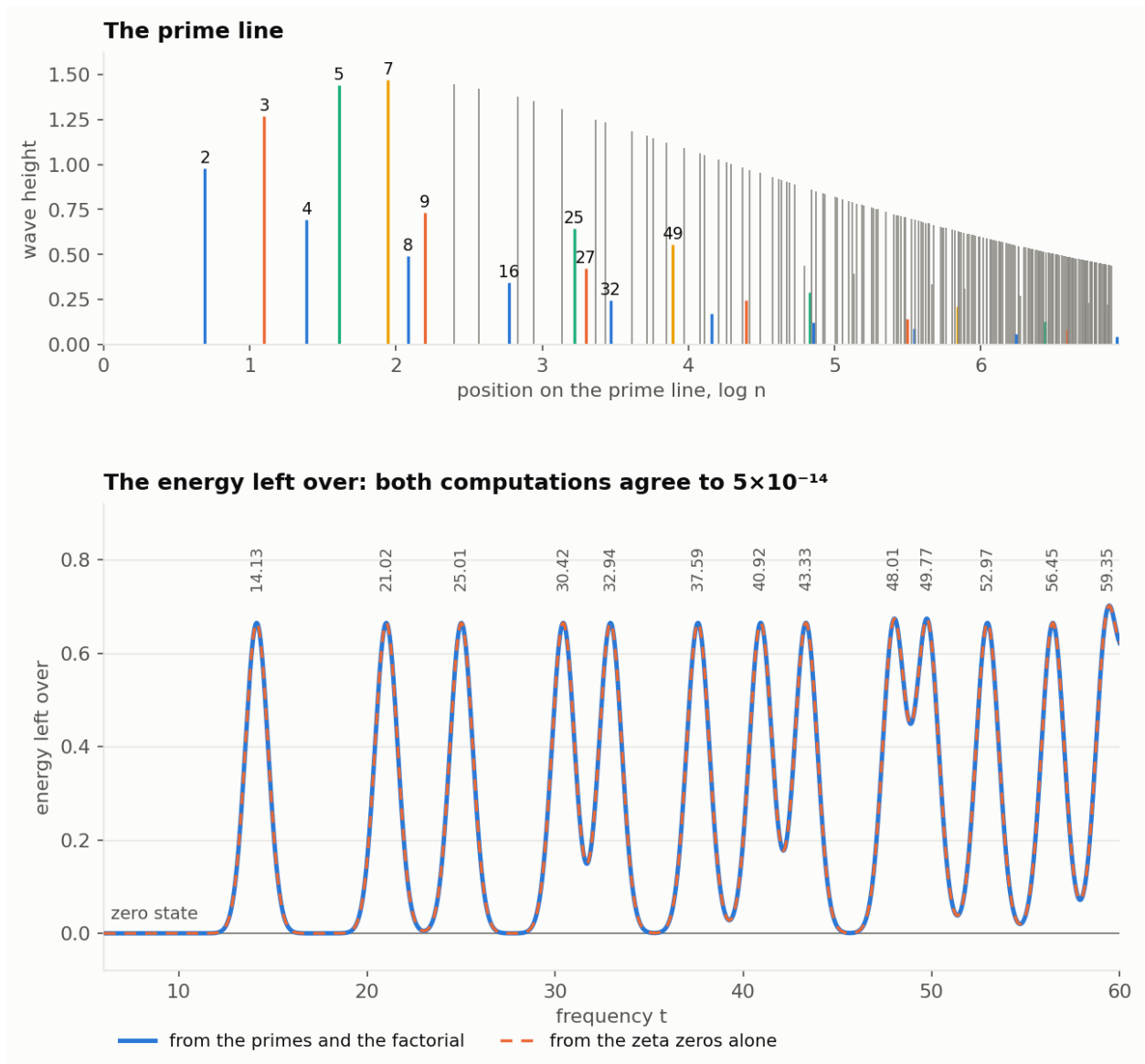


FIGURE 5. Top: the prime line, every prime power at $\log n$ with height $2\Lambda(n)/\sqrt{n}$. Bottom: the archimedean energy minus all 78,734 prime-power waves up to 10^6 (solid) against Gaussian bumps placed at the zeta zeros alone (dashed); the two agree to 5×10^{-14} .

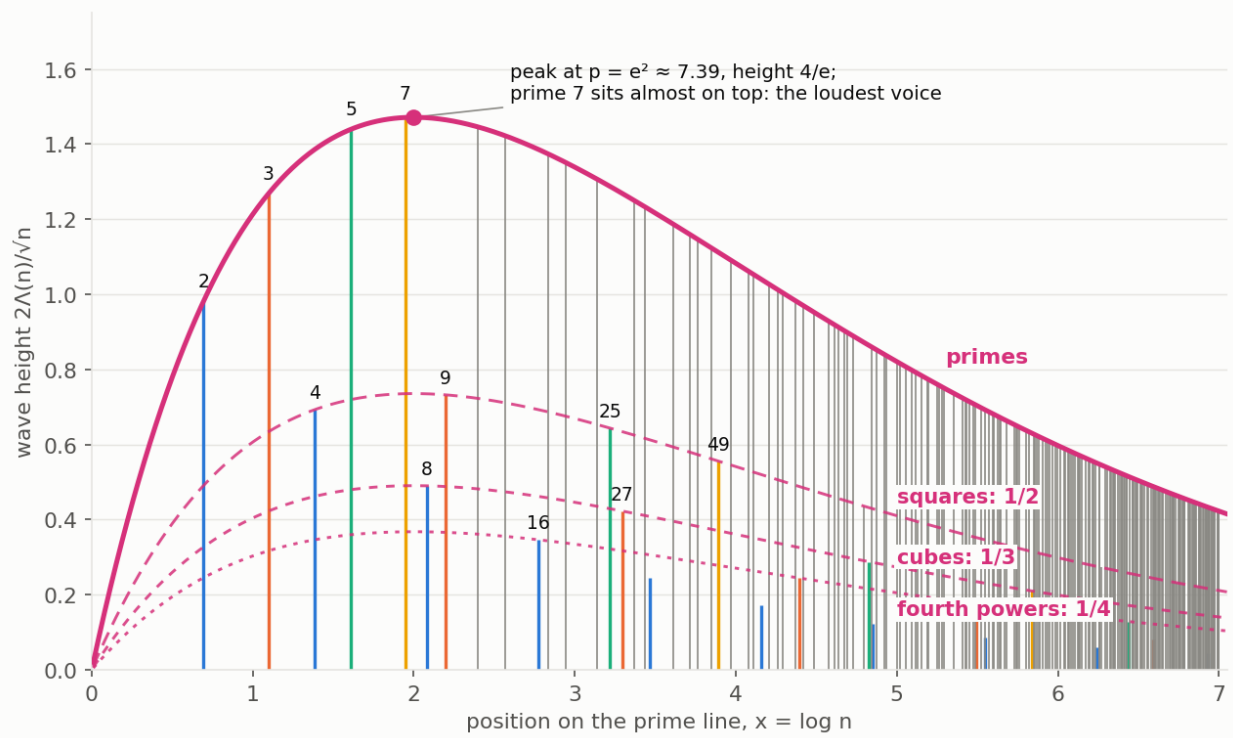


FIGURE 6. The envelope of Proposition 4.1: primes on $2xe^{-x/2}$, squares on $1/2$ of it, cubes on $1/3$, fourth powers on $1/4$. The prime 7 sits almost at the peak $p = e^2$.

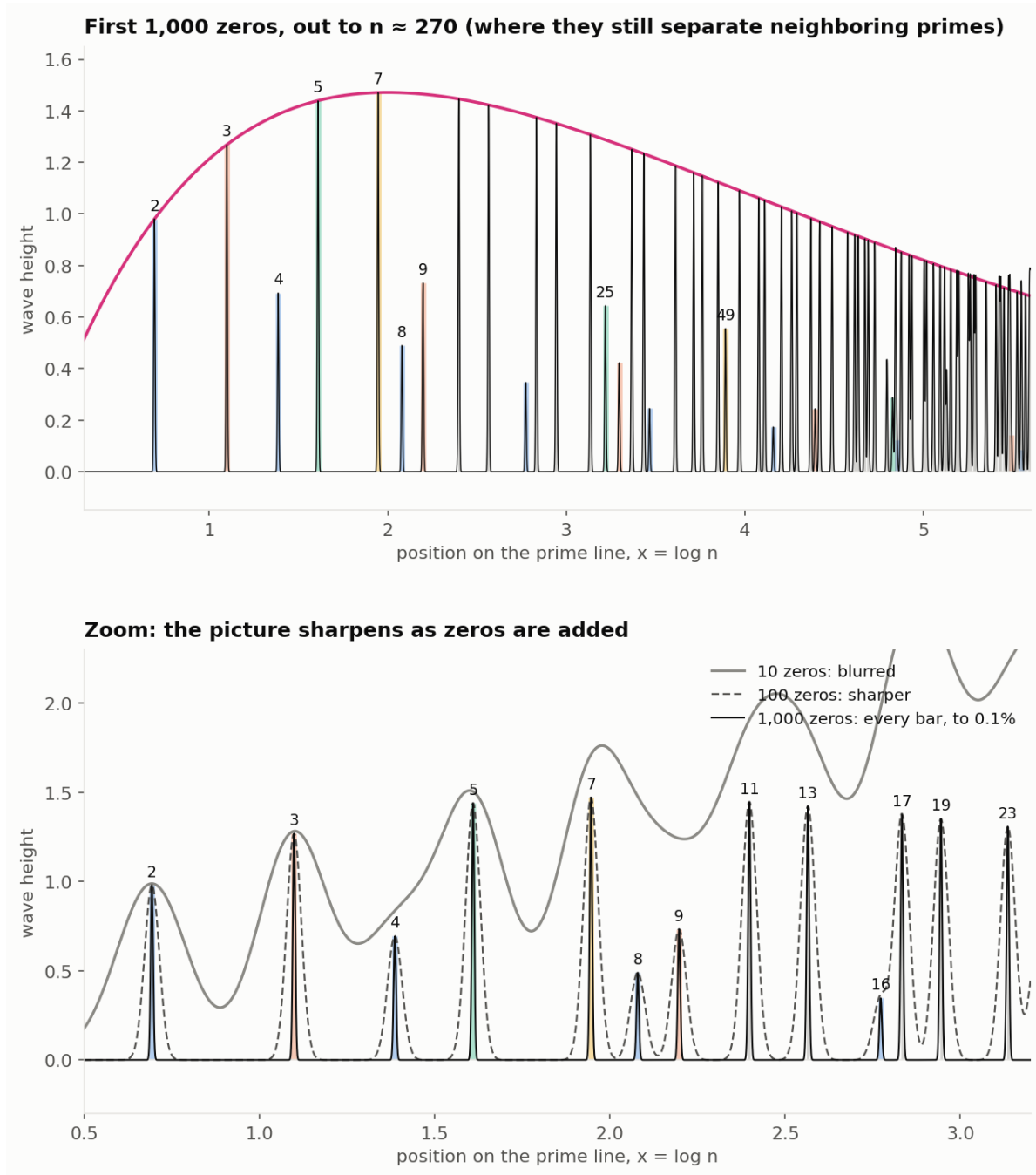


FIGURE 7. The zeros rebuild the prime line. Black: a curve built only from zeta zeros (the smooth term minus one cosine wave per zero). With 1,000 zeros every bar out to $n \approx 270$ is matched to 0.1%; the lower panel shows the sharpening from 10 to 100 to 1,000 zeros.

5. THE PARABOLA SIEVE

The Matiyasevich–Stechkin sieve [26] places the points (a^2, a) and $(b^2, -b)$ on the parabola $x = y^2$ and joins them by chords.

Proposition 5.1. *For integers $a, b \geq 1$ the chord through (a^2, a) and $(b^2, -b)$ is the line $x = ab + (a - b)y$. It meets the axis $y = 0$ at $x = ab$, and it makes the angle $\arctan(1/|a - b|)$ with the axis. Consequently every composite number is the axis crossing of a chord with $a, b \geq 2$, and no prime is.*

Proof. At $y = a$ the line gives $x = ab + a^2 - ab = a^2$, and at $y = -b$ it gives $x = ab - ab + b^2 = b^2$. The remaining claims are immediate. $\square$

In the figures we keep only chords whose top endpoint is a prime, (p^2, p) , and color each chord by the prime factors of its endpoints. Every composite $n \leq N$ is then reached once from each of its prime factors p (with partner $b = n/p$).

Computation 5.2. Up to 49 the rule of 2 reaches 23 numbers, the rule of 3 reaches 15, the rule of 5 reaches 8 and the rule of 7 reaches 6. The numbers of *new* composites (not reached by a smaller prime) are 23, 7, 2 and 1, for a total of 33, the number of composites in $[4, 49]$ (Figures 8–11).

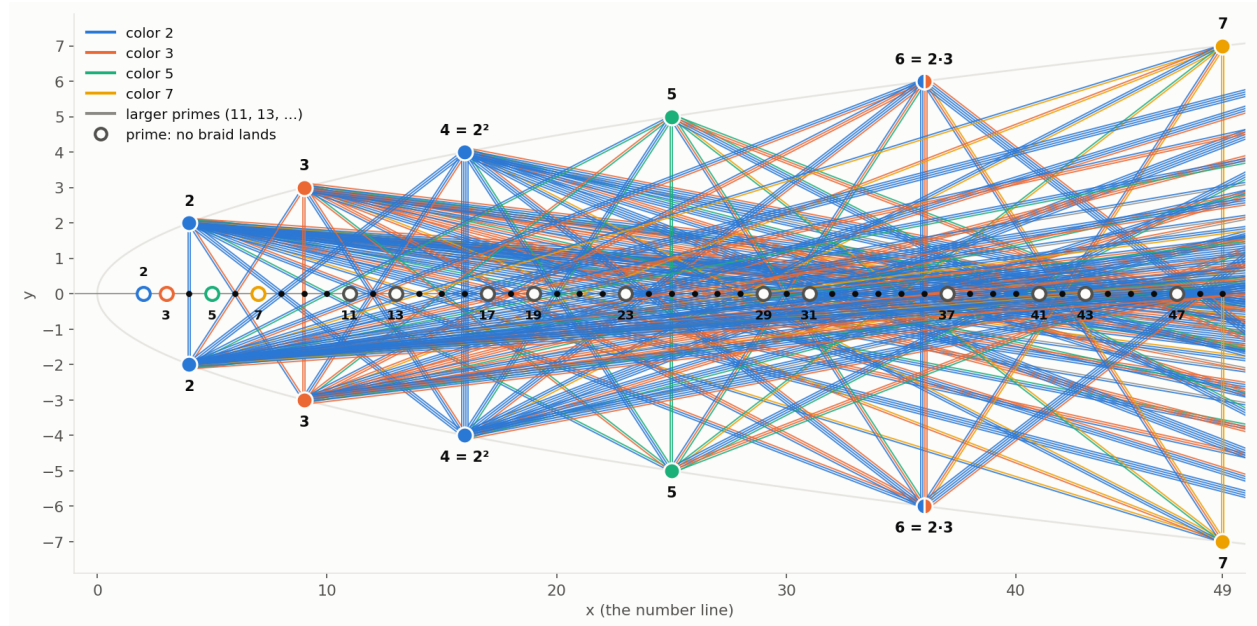


FIGURE 8. The parabola sieve drawn as braids: one strand for each prime factor of ab , in that prime's color, so that multiplication adds color vectors. Open circles on the axis are the primes.

5.1. The upper edge. Let $\mathcal{C}$ be the set of chords from (r^2, r) , r prime, to $(b^2, -b)$, $b \geq 2$ an integer, $b \neq r$, and let $E(x)$ be the supremum of the heights $y \geq 0$ of the chords in $\mathcal{C}$ above the point x . The top half of the picture is bounded by the graph of E .

Proposition 5.3 (Staircase). *Let $p < q$ be consecutive primes and $g = q - p$. For $p^2 \leq x \leq q^2$,*

$$E(x) = \max\left\{p, \frac{x - 2q}{q - 2}\right\}.$$

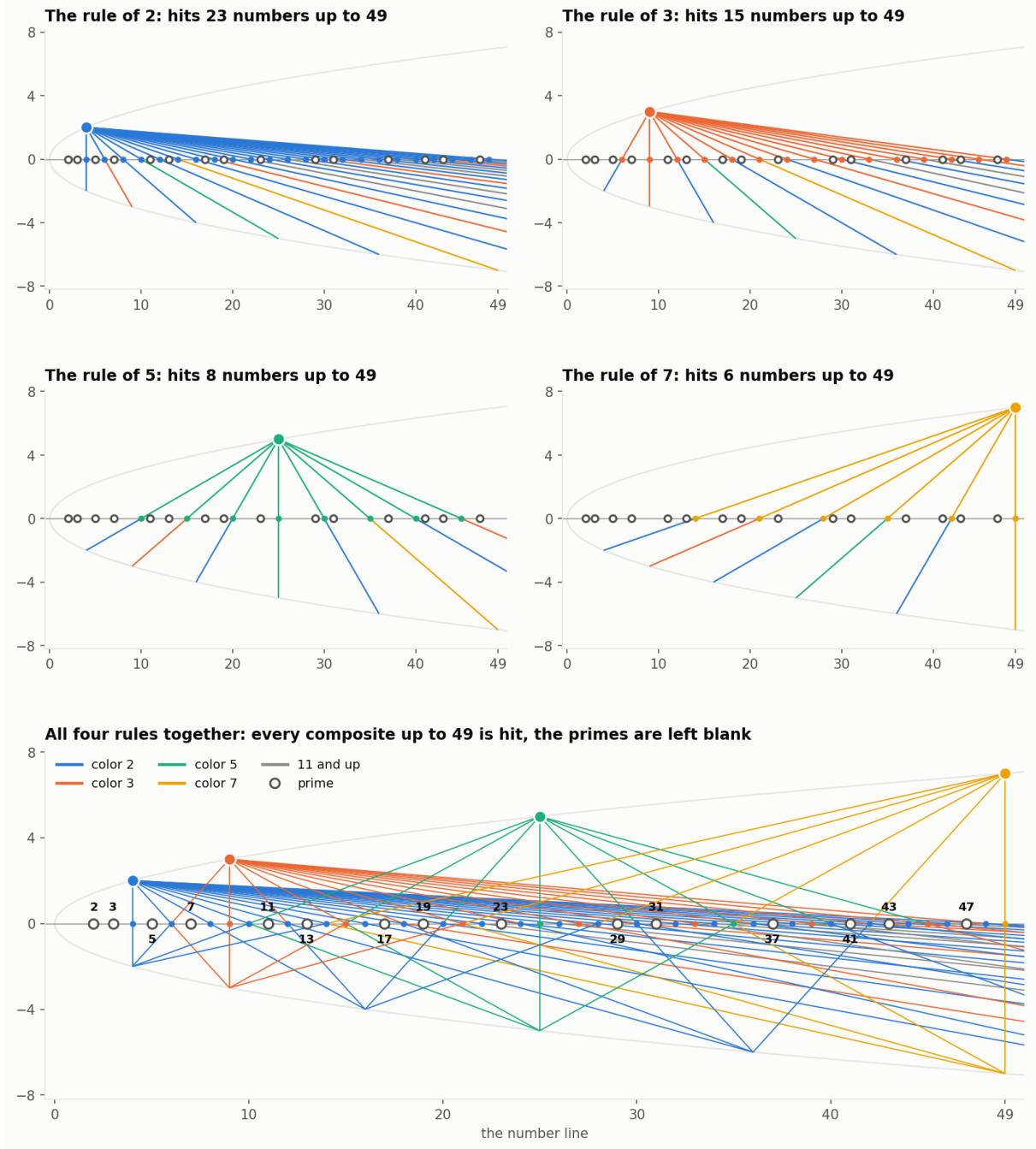


FIGURE 9. One rule per panel: chords from (p^2, p) to every partner $(b^2, -b)$, colored by the prime on top (upper half) and by the smallest prime of the partner (lower half). Together the four rules reach every composite up to 49 and leave the primes unreached.

Thus E is flat at height p on an interval of length $(q-p)(p+2)$ and then rises along the chord from (q^2, q) to $(4, -2)$ on an interval of length $(q-p)(q-2)$. The two lengths differ by $g(4-g)$: a gap of 2 makes the flat longer by 4, a gap of 4 makes them equal, and a gap of 6 or more makes the ramp longer.

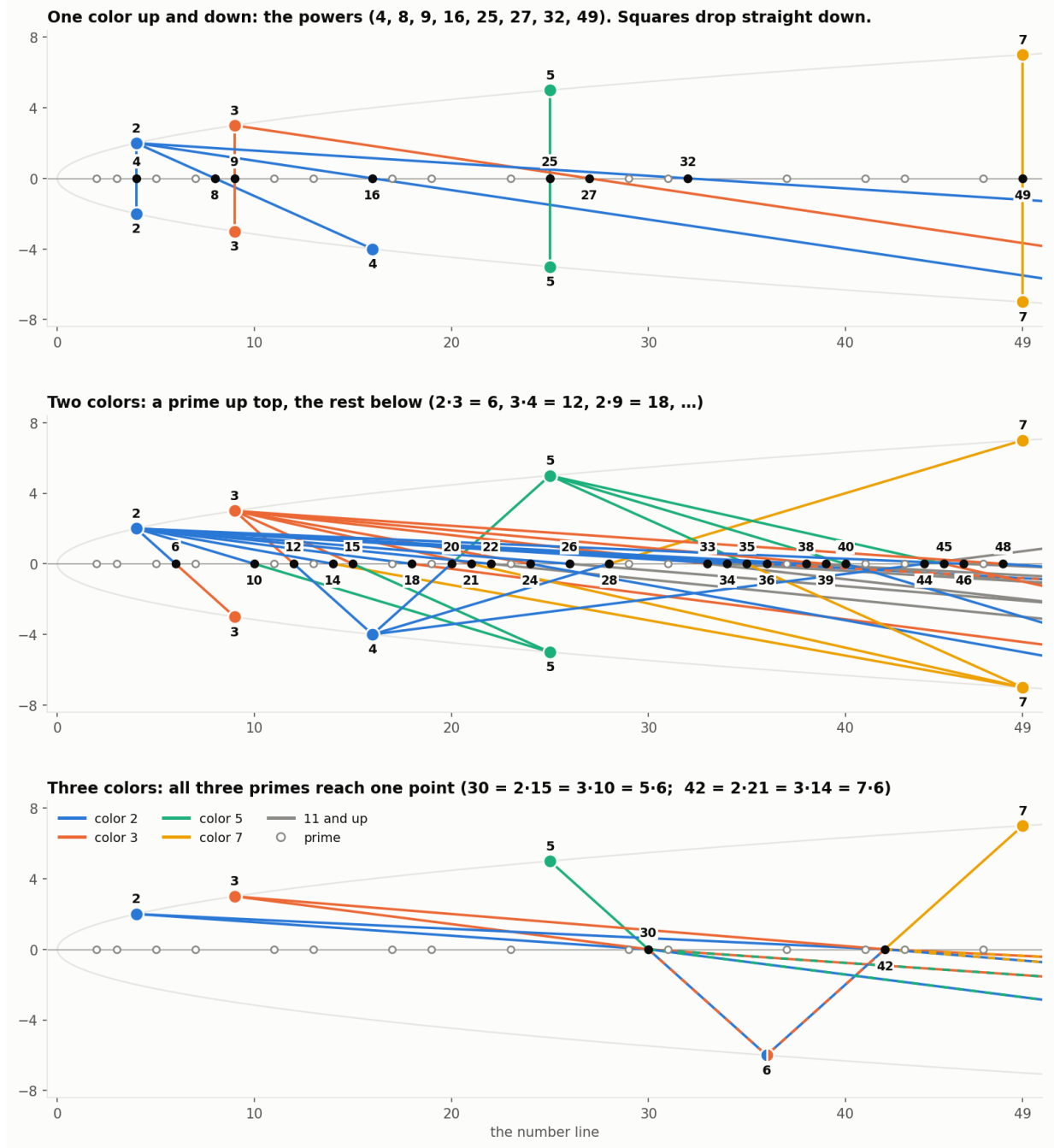


FIGURE 10. Chords sorted by type. Top: prime powers (4, 8, 9, 16, 25, 27, 32, 49); squares drop vertically. Middle: two colors, a prime on top and a partner below. Bottom: three colors meeting at one point (30 = 2 · 15 = 3 · 10 = 5 · 6 and 42 = 2 · 21 = 3 · 14 = 7 · 6).

Proof. Write $h_{r,b}(x) = (x - rb)/(r - b)$ for the height above x of the chord from (r^2, r) to $(b^2, -b)$, which by Proposition 5.1 is nonnegative between $x = rb$ and $x = r^2$. If $b > r$ the chord runs from (r^2, r) down to $(rb, 0)$ and $h_{r,b}(x) = r - (x - r^2)/(b - r) < r$ for $x > r^2$; letting $b \rightarrow \infty$ shows that the chords of r contribute exactly $\sup = r$ to $E(x)$ for every $x \geq r^2$. For $x \leq q^2$ the chords of primes $r \geq q$ with $b > r$ lie to the right of x . If $b < r$ we have $h_{r,b}(x) - r = (x - r^2)/(r - b)$, which for

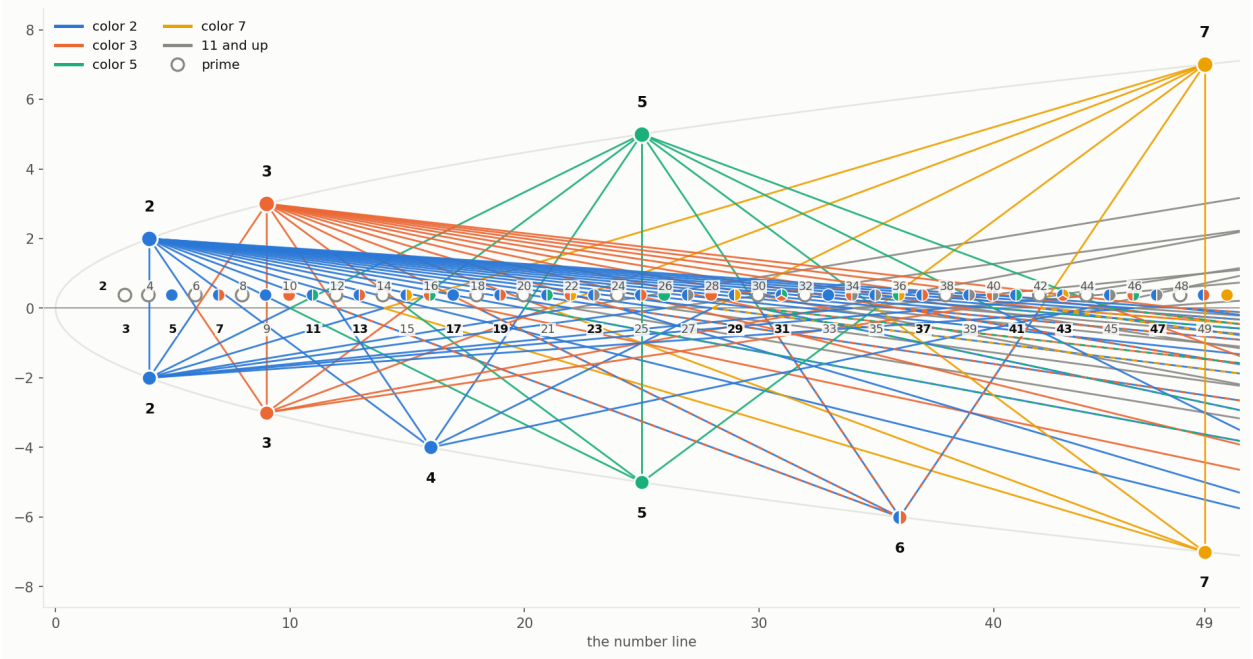


FIGURE 11. The full sieve with primes on top: 60 chords, every composite up to 49 reached once from each of its prime factors. Axis markers carry each number's own colors; primes are open gray circles.

$x < r^2$ is largest when $r - b$ is largest, that is for $b = 2$. Finally, for $r > q$ and $x > 4$,

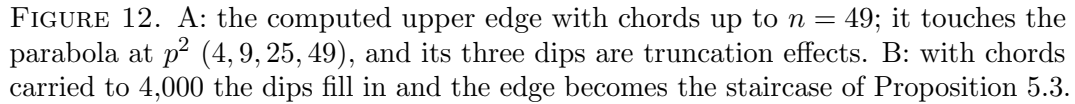
$$\frac{x - 2r}{r - 2} < \frac{x - 2q}{q - 2} \iff 4(r - q) < x(r - q),$$

so on $[p^2, q^2]$ the chord $h_{q,2}$ dominates every chord of a larger prime. Collecting terms, $E(x) = \max\{\max_{r \leq p} r, h_{q,2}(x)\} = \max\{p, (x - 2q)/(q - 2)\}$. The ramp meets height p at $x_0 = p(q - 2) + 2q = p^2 + (q - p)(p + 2)$, so the flat has length $(q - p)(p + 2)$ and the ramp $q^2 - x_0 = (q - p)(q - 2)$. Their difference is $(q - p)(p + 4 - q) = g(4 - g)$. $\square$

TABLE 3. The staircase of Proposition 5.3 for the first primes.

$p \rightarrow q$	Gap	Flat	Length	Ramp	Length
$2 \rightarrow 3$	1	$[4, 8]$	4	$[8, 9]$	1
$3 \rightarrow 5$	2	$[9, 19]$	10	$[19, 25]$	6
$5 \rightarrow 7$	2	$[25, 39]$	14	$[39, 49]$	10
$7 \rightarrow 11$	4	$[49, 85]$	36	$[85, 121]$	36
$11 \rightarrow 13$	2	$[121, 147]$	26	$[147, 169]$	22
$13 \rightarrow 17$	4	$[169, 229]$	60	$[229, 289]$	60

Computation 5.4. With chords to all partners up to 4,000 the measured upper edge agrees with Proposition 5.3 to within 0.072 on $[4, 180]$. With chords only to partners up to 49 the edge shows three extra dips, at $x = 7.83, 17.13$ and 31.22 (heights 1.83, 2.38, 3.44), which are truncation effects (Figure 12).

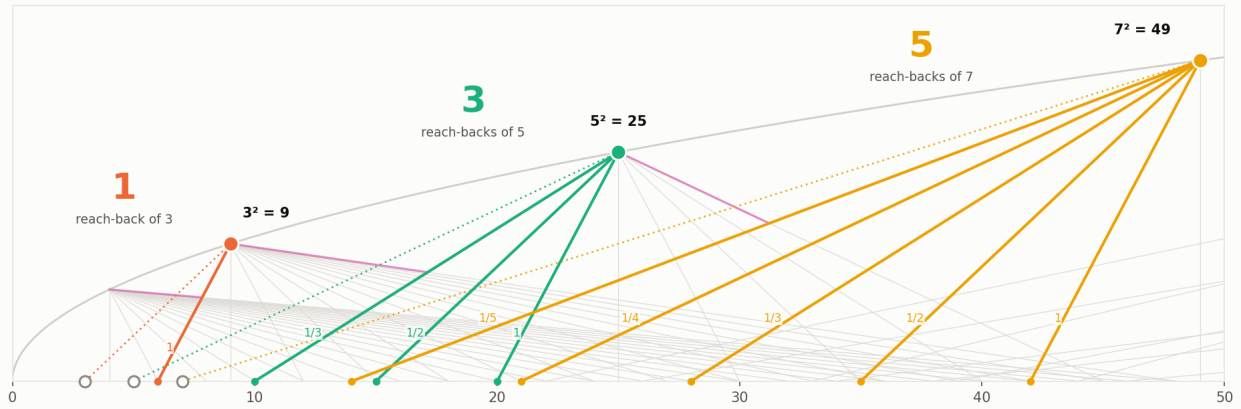


5.2. Reach-backs.

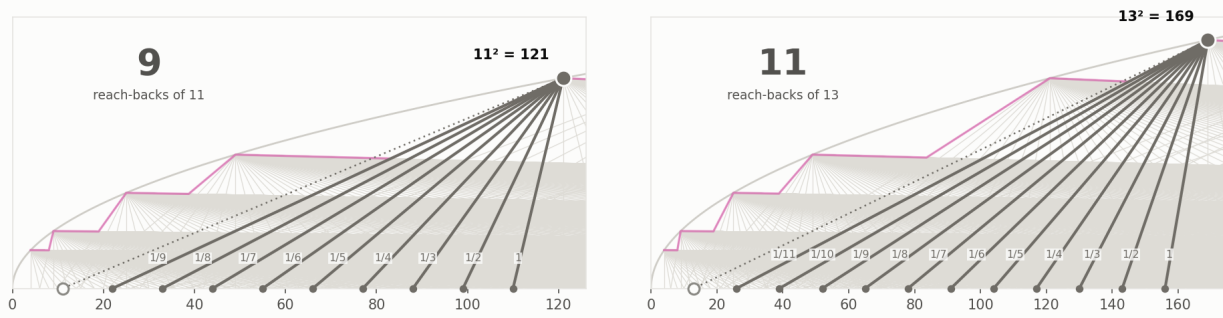
Proof. Everything follows from Proposition 5.1; for the ramp, compare with the proof of Proposition 5.3. $\square$

Remark 5.7. The reach-back picture reads backward in time (the square q^2 “reaches into” multiples below it), but it records the primality of q rather than predicting it: to draw the reach-backs

A. The window to 49. The counts are $q - 2$. Magenta (faint) is the upper edge of Figure 12: each ramp is the $1/(q - 2)$ reach-back.



B, C. The next two primes. The count is 9, then 11, not 7: the reach-back count is $q - 2$, so it only takes values two below a prime.

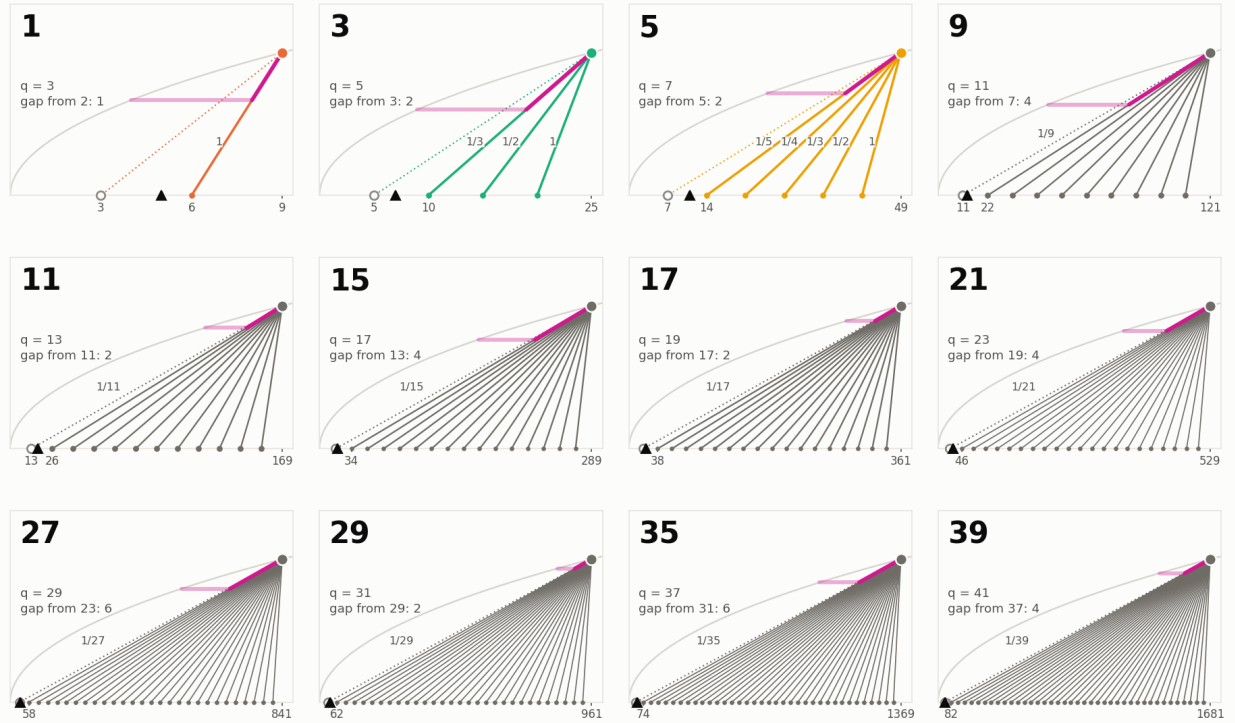


Reach-back counts: 1, 3, 5, 9, 11, 15, 17, 21, 27, 29, ... (every prime minus 2)

Sieve of Eratosthenes starts each prime at q^2 for exactly this reason: the $q - 2$ reach-back hits are already done. They add no new records. They only set the angles.

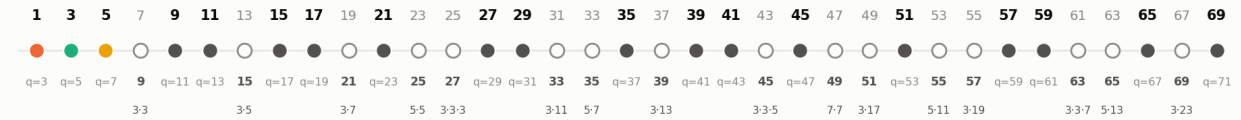
FIGURE 13. Reach-backs. A: in the window up to 49 the reach-back counts of 3, 5 and 7 are 1, 3 and 5, with slopes labelled; the dotted segment from (q^2, q) to q is the chord that is never drawn. B, C: the next primes 11 and 13.

one must already know that q is prime. The staircase encodes every prime and every gap, so extrapolating its next step without knowing the next prime would amount to a formula for the primes.



Why the count skips 7: an odd number k is a count only when $k + 2$ is prime.

Filled: a count ($k + 2$ is prime). Open: a hole, with the composite $k + 2$ written under it. 7 is missing because $9 = 3 \cdot 3$, which is rule 3's first new record.



Holes sit at 7, 13, 19, 23, 25, 31, 33, 37, 43, 47, ... : two below each odd composite. So the rule counts carry the primes; they do not generate them.

FIGURE 14. Every rule from 3 to 41, with the count $q - 2$, the ramp, the un-drawn chord and the next prime (triangle). Bottom: the count ladder; the holes 7, 13, 19, 23, 25, 31, 33, ... sit two below odd composites.

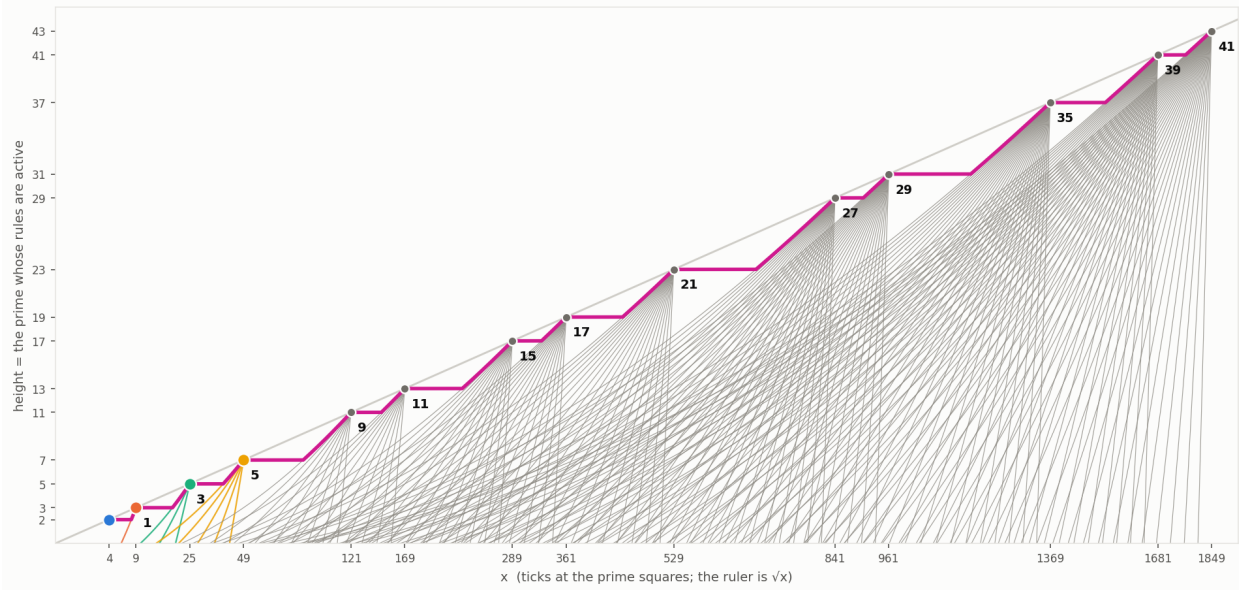


FIGURE 15. All rules on a $\sqrt{x}$ ruler, on which the parabola becomes the diagonal and each prime sits at (q, q) . The staircase climbs the diagonal: a flat step at each prime, then a ramp that is the next prime's shallowest reach-back.

6. PRIME GAPS TO 10^{11} AS A CHAIN OF TRIANGLES

On the $\sqrt{x}$ ruler $X = \sqrt{x}$, write $D(X) = X - E(X^2)$ for the vertical distance between the diagonal and the staircase. Between consecutive primes $p < q$ the region under D is a “tooth”. The sequence of teeth is called the *magenta walk* in [28], after its color in the figures.

Proposition 6.1 (Teeth). *Let $p < q$ be consecutive primes with gap g , and let $X_0 = \sqrt{p^2 + g(p+2)}$. Then*

$$D(X) = \begin{cases} X - p, & p \leq X \leq X_0, \\ X - \frac{X^2 - 2q}{q - 2}, & X_0 \leq X \leq q, \end{cases}$$

so $D(p) = D(q) = 0$, D rises with slope 1 and falls with slope between -1 and $-(q+2)/(q-2)$. The peak lies at $X_0 - p = \frac{g}{2}(1 + \frac{2}{p}) + O(g^2/p)$. Consequently, after rescaling both axes by g , the tooth converges uniformly to the triangle with vertices $(0, 0)$, $(\frac{1}{2}, \frac{1}{2})$, $(1, 0)$ as $g/p \rightarrow 0$.

Proof. Substitute $x = X^2$ into Proposition 5.3. The slope of the second branch is $1 - 2X/(q-2)$, which lies between $-1 - 4/(q-2)$ and $-1 + O(g/q)$ on $[X_0, q]$, and its curvature is $-2/(q-2)$. Expanding $X_0 = p\sqrt{1 + g(p+2)/p^2}$ gives the peak position. $\square$

Proposition 6.2 (Residue alternation). *Every prime $p > 3$ is $\equiv \pm 1 \pmod{6}$. A gap $g \equiv 2 \pmod{6}$ moves from the class -1 to $+1$, a gap $g \equiv 4 \pmod{6}$ moves from $+1$ to -1 , and a gap $g \equiv 0 \pmod{6}$ stays in the same class. Hence, after discarding the gaps divisible by 6, the remaining gaps are alternately $\equiv 2$ and $\equiv 4 \pmod{6}$, and the tooth heights $g/2$ not divisible by 3 alternate between 1 and 2 $\pmod{3}$.*

Proof. Immediate from $p \equiv \pm 1 \pmod{6}$. $\square$

Computation 6.3 (All gaps to 10^{11}). A segmented sieve (Appendix A) reproduced $\pi(10^{11}) = 4,118,054,813$ and all 40 record gaps below 10^{11} in the published tables [32, 44]. Among the 4,118,054,810 gaps between primes $5 \leq p < q \leq 10^{11}$ the alternation of Proposition 6.2 holds without exception, and the transitions from the class 1 $\pmod{6}$ to the class 5 $\pmod{6}$ and back both number 1,124,275,248.

Computation 6.4 (Hardy–Littlewood weights). For $10^{10} \leq p \leq 10^{11}$ the frequency of each gap g , relative to a smooth baseline, matches the singular-series factor $\prod_{r|g, r>2} (r-1)/(r-2)$ of the Hardy–Littlewood prime-pair conjecture [21]: $g = 6$ occurs 1.94 times the baseline (predicted 2), $g = 30$ occurs 2.68 times (predicted 2.67) and $g = 210$ occurs 3.20 times (predicted 3.20). The median deviation over $g \leq 300$ is 2.2%.

Theorem 6.5 (Carneiro, Milinovich and Soundararajan [8]). *Assume RH. Then for every prime $p > 3$ the gap to the next prime satisfies $q - p < \frac{22}{25}\sqrt{p} \log p$.*

Computation 6.6. Table 4 compares the gaps with the ceiling of Theorem 6.5 and with the Cramér scale $\log^2 p$ [14], using our data to 10^{11} and the published maximal gaps to about 10^{20} [44].

Remark 6.7. The walk stays far below the RH ceiling and close to the Cramér scale. RH caps the teeth, but proving that they stay on the Cramér scale would be a much stronger statement than RH.

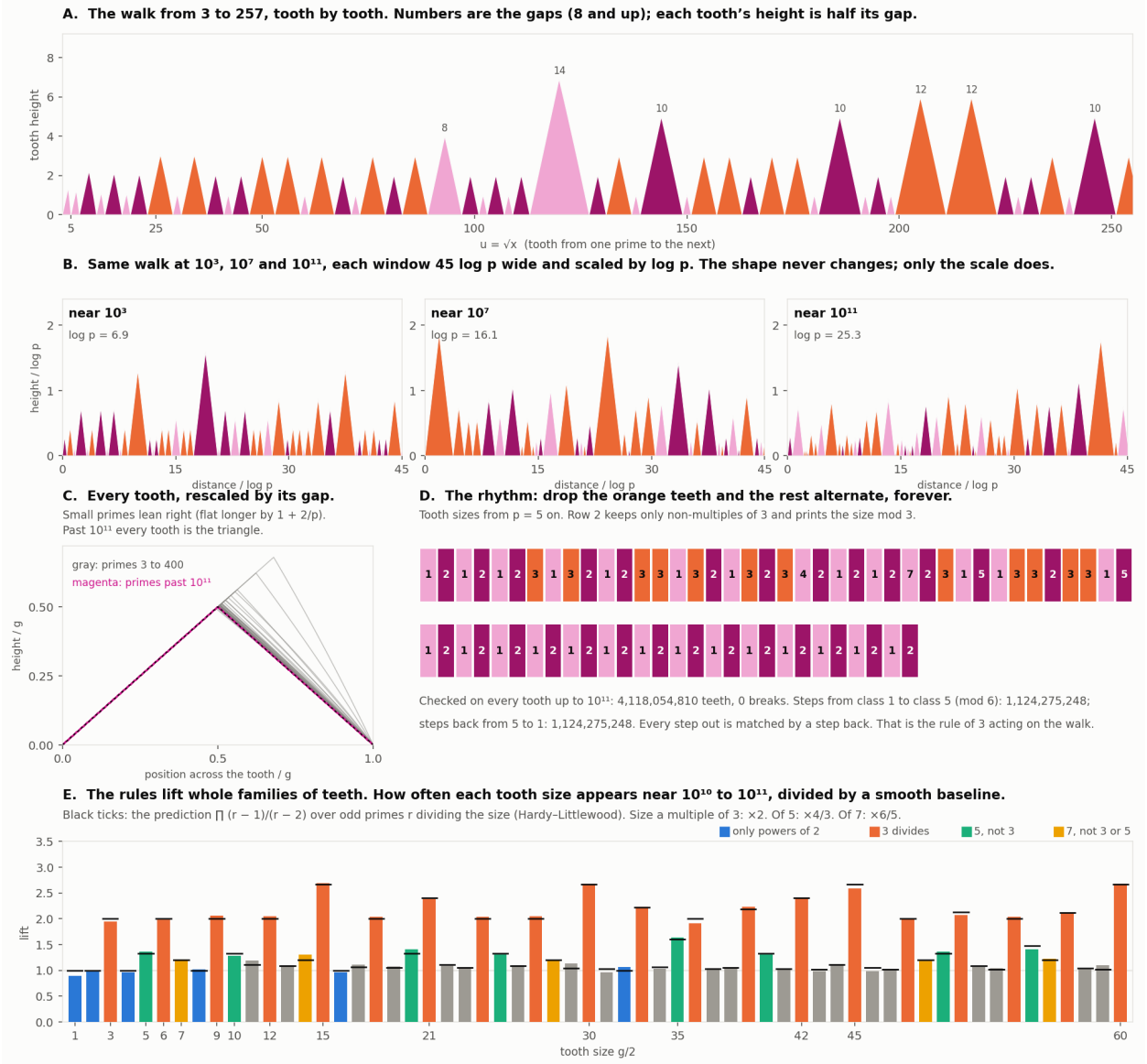


FIGURE 16. A: the walk from 3 to 257 tooth by tooth (labels are gaps of 8 and more). B: the walk near 10^3 , 10^7 and 10^{11} in windows of $45 \log p$, scaled by $\log p$; the shape does not change with scale. C: every tooth rescaled by its gap collapses onto one triangle past 10^{11} (Proposition 6.1). D: the residue alternation of Proposition 6.2. E: tooth frequencies near 10^{10} to 10^{11} divided by a smooth baseline, against the Hardy-Littlewood factor (Computation 6.4).

TABLE 4. The walk against its ceilings.

Quantity	Value	Status
Room between the largest tooth and the RH ceiling at 10^{20}	factor 2.2×10^8	computed
Largest $(q - p)/\log^2 p$ over record gaps with $p > 7$	0.9206 (gap 1132 after 1,693,182,318,746,371)	computed from [44]
Record gaps below $\log^2 p$ for $p > 7$	all 85 published records	consistent with Cramér's conjecture
Tooth area $\sum_{p \leq X} g^2/4$ divided by $X \log X$	0.261 at 10, 0.337 at 10^4 , 0.407 at 10^8 , 0.429 at 10^{11}	computed; limit 1/2 conjectural
Bound on $\sum_{p \leq X} g^2$ under RH	$O(X \log^3 X)$	Selberg [36]

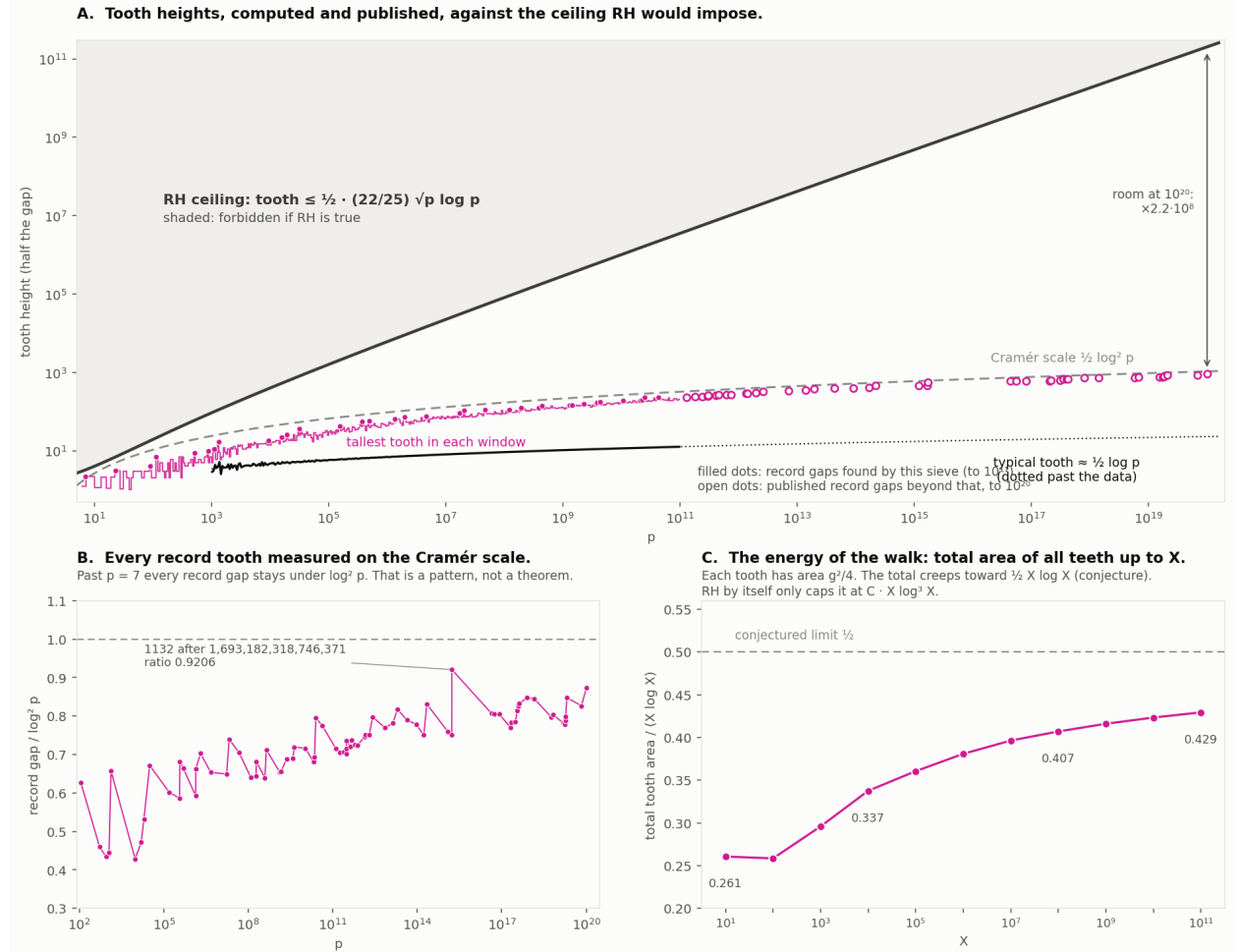


FIGURE 17. A: tooth heights computed to 10^{11} (filled dots, tallest tooth in each window) and published record gaps to 10^{20} (open dots), against the ceiling of Theorem 6.5 (shaded region is excluded if RH holds). B: record teeth on the Cramér scale $\log^2 p$. C: total tooth area up to X divided by $X \log X$.

7. DERIVATIVE JETS OF ξ

For $z \in \mathbb{C}$ let $c(z) = (c_j(z))_{j \geq 0}$ with $c_j(z) = F^{(j)}(z)/j!$, the Taylor jet of $F(z) = \xi(\frac{1}{2} + z)$ at z , and let $G(z) = c(z)c(z)^*$ be the matrix of all products $c_j \bar{c}_k$. (In [28] these are called the infinity plane and the quadratic plane.) Since F is entire of order 1, the jet decays faster than any geometric sequence.

Proposition 7.1. *Let $(T_h)_{jk} = \binom{k}{j} h^{k-j}$ for $k \geq j$. Then $c(z+h) = T_h c(z)$ and $G(z+h) = T_h G(z) T_h^*$; G is Hermitian, positive semidefinite and of rank one. Writing $z = a + it$,*

$$K := \operatorname{Re} G_{10} = \operatorname{Re}(F'(z) \overline{F'(z)}) = \frac{1}{2} \partial_a |F|^2 = |F|^2 \partial_t \arg F.$$

Proof. The first identity is Taylor's theorem, and the second follows. For the last line, $\partial_a |F|^2 = 2 \operatorname{Re}(\bar{F} \partial_a F) = 2 \operatorname{Re}(\bar{F} F')$, and $\partial_t \arg F = \operatorname{Im}(\partial_t \log F) = \operatorname{Im}(iF'/F) = \operatorname{Re}(F'/F)$. $\square$

Theorem 7.2 (Sondow and Dumitrescu [38]; see also [27]). *RH is equivalent to the horizontal monotonicity of $|\xi|$ in the half-plane $\operatorname{Re} s > \frac{1}{2}$, that is, to $K \geq 0$ for $a > 0$.*

Computation 7.3. On the real ξ at 40-digit precision (Table 5, Figure 18): the jet at $z = 0.3 + 14i$ rebuilds F a full unit away to 40 digits; the quantity $\Omega = K/|F|^2$ is positive on all 38,400 grid points with $0.002 \leq a \leq 1.5$ and $0 < t \leq 60$, with smallest value 9×10^{-5} . Multiplying ξ by a polynomial factor that plants one off-line zero at $a = 0.2$, $t = 35.5$ produces a region where $K < 0$.

 TABLE 5. Checks of Proposition 7.1 on the real ξ .

Identity	Result
$c(z+h) = T_h c(z)$	agreement to 37 digits
every 2×2 minor of G vanishes	to 5×10^{-16}
rank one	one eigenvalue $\ c\ ^2$, the rest about 10^{-22}
$G(z+h) = T_h G T_h^*$	to 2.5×10^{-16}
c recovered from one column of G	to 7×10^{-17} , up to phase
three expressions for K	all equal $6.7857572987 \times 10^{-7}$

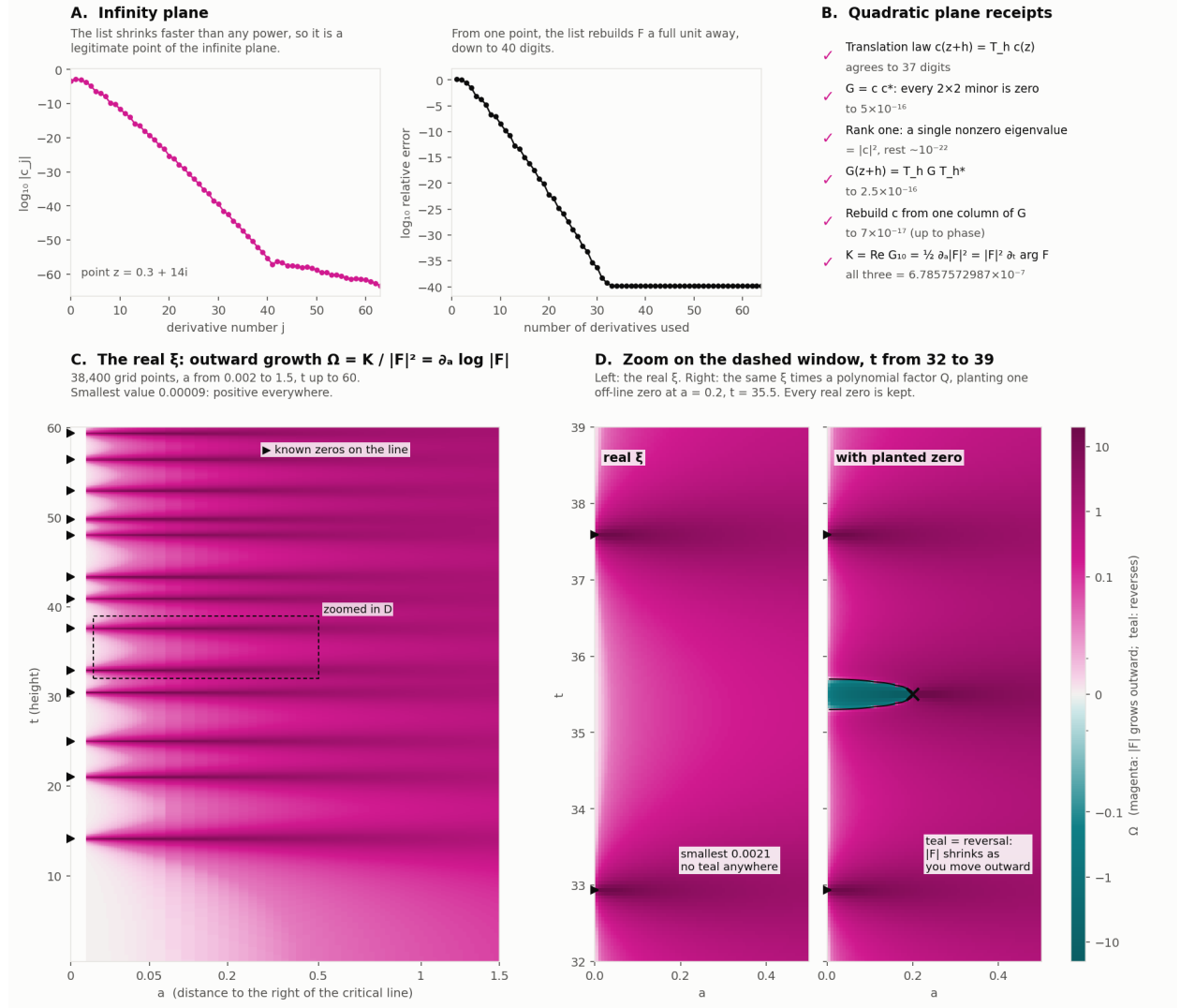


FIGURE 18. A: the jet at $z = 0.3 + 14i$ and its reconstruction error. B: the checks of Table 5. C: $\Omega = K/|F|^2$ over the grid of Computation 7.3, positive everywhere. D: a zoom ($32 \leq t \leq 39$) of the real ξ beside ξ multiplied by a polynomial factor planting one off-line zero at $a = 0.2, t = 35.5$; a region with $K < 0$ (teal) appears.

8. THE FUNCTIONAL-EQUATION FOLD AND THE DAVENPORT–HEILBRONN CONTROL

Proposition 8.1. $F(-z) = F(z)$ and $F(\bar{z}) = \overline{F(z)}$. Consequently $|F(-a + it)| = |F(a + it)|$, so $K = 0$ on the critical line; and on the critical line the jet satisfies $c_j(it) \in \mathbb{R}$ for even j and $c_j(it) \in i\mathbb{R}$ for odd j , so that $G(it)$ has a checkerboard pattern of real and imaginary entries and $\text{Re } G_{10}(it) = 0$.

Proof. The symmetries are the functional equation $\xi(s) = \xi(1-s)$ and the reflection principle. The function $\Xi(t) = F(it)$ is real for real t , and $F^{(j)}(it) = (-i)^j \Xi^{(j)}(t)$. $\square$

Computation 8.2. The mirror symmetry holds numerically to 5×10^{-15} . On the line ($z = 20i$) $\text{Re } G_{10}$ is about 10^{-50} against entries of size 10^{-9} ; at $z = 0.3 + 20i$ the checkerboard breaks and $\text{Re } G_{10} = 4.4 \times 10^{-10}$ (Figure 19).

The symmetries of Proposition 8.1 fix the critical line as a line of symmetry but do not force the zeros onto it. The standard witness is the Davenport–Heilbronn function [16], a linear combination of two Dirichlet L -functions modulo 5 that satisfies a functional equation of the same shape, has no Euler product, and has zeros off the critical line.

Computation 8.3. Using a Hurwitz-zeta representation we reproduce the known off-line zero of the Davenport–Heilbronn function at $0.80851718245 + 85.69934848538i$ [3] and find four off-line pairs up to height 200, at heights 85.7, 114.2, 166.5 and 176.7, among 130 zeros in total (the smooth count predicts 129.5). Around each off-line pair there is a region where $|f|$ decreases away from the critical line (Figures 19–21).

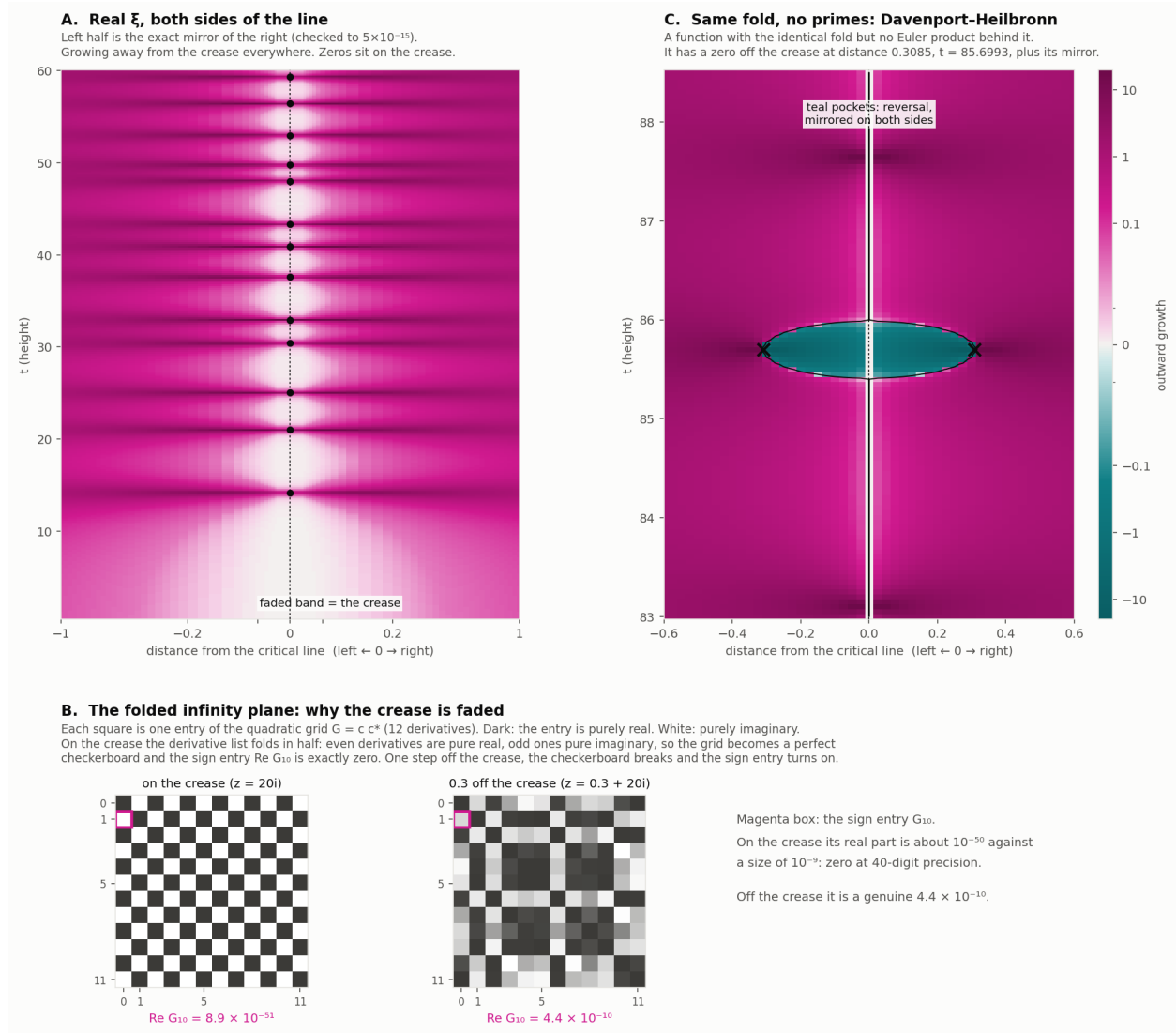


FIGURE 19. A: Ω for the real ξ on both sides of the critical line; the left half mirrors the right and the growth vanishes on the line. B: the jet grid G on the line (a perfect checkerboard of real and imaginary entries) and 0.3 off it. C: the Davenport–Heilbronn function, with the same symmetry, an off-line zero at $0.8085 + 85.6993i$ and its mirror, and regions of reversed growth (teal).

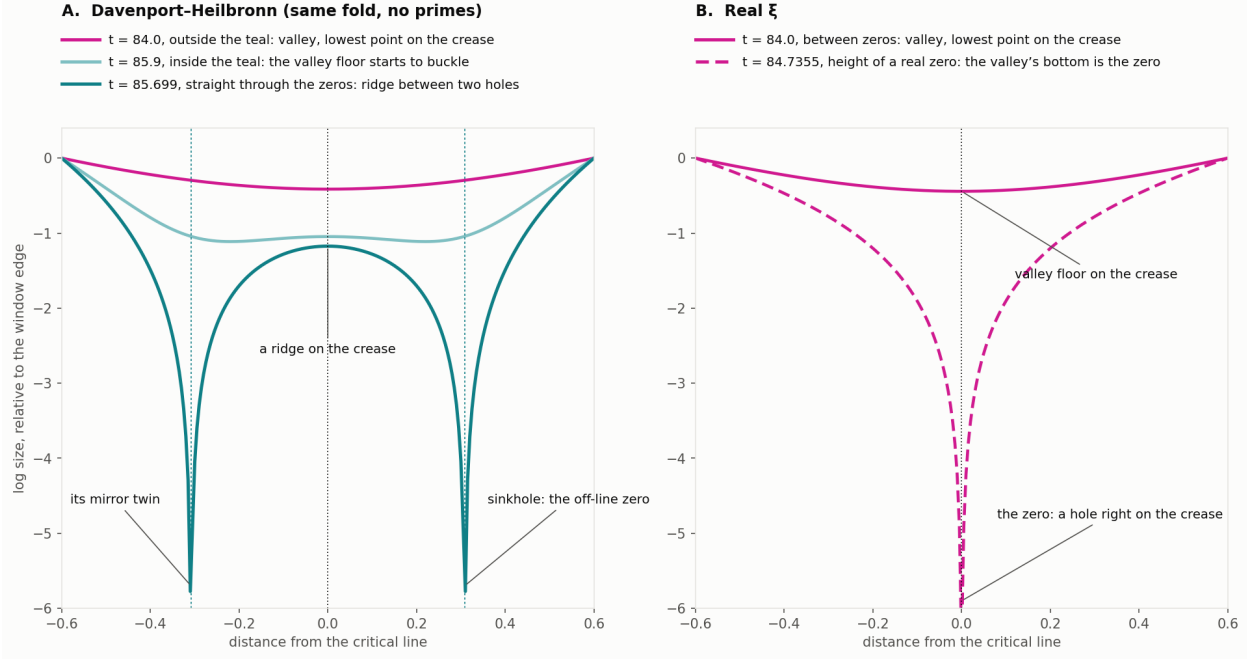


FIGURE 20. Horizontal slices $|f(\sigma + it)|$ as functions of σ . Right, real ξ : every slice is a valley with its minimum on the critical line, and a zero is a point where the valley floor reaches 0. Left, Davenport–Heilbronn: inside the reversal region the critical line becomes a ridge between two holes, the off-line zero and its mirror.

Remark 8.4. What ξ has and the Davenport–Heilbronn function lacks is the Euler product. An off-line quartet raises the zero count without adding sign changes of Ξ on the line; this is the basis of Turing’s method [40], by which all zeros up to height 3×10^{12} are known to lie on the line [33]. Folding once more by $w = z^2$ turns RH into the statement that all zeros of $F(\sqrt{w})$ are real and negative, the setting of Pólya’s program and of the de Bruijn–Newman constant, which Rodgers and Tao proved to be nonnegative [35].

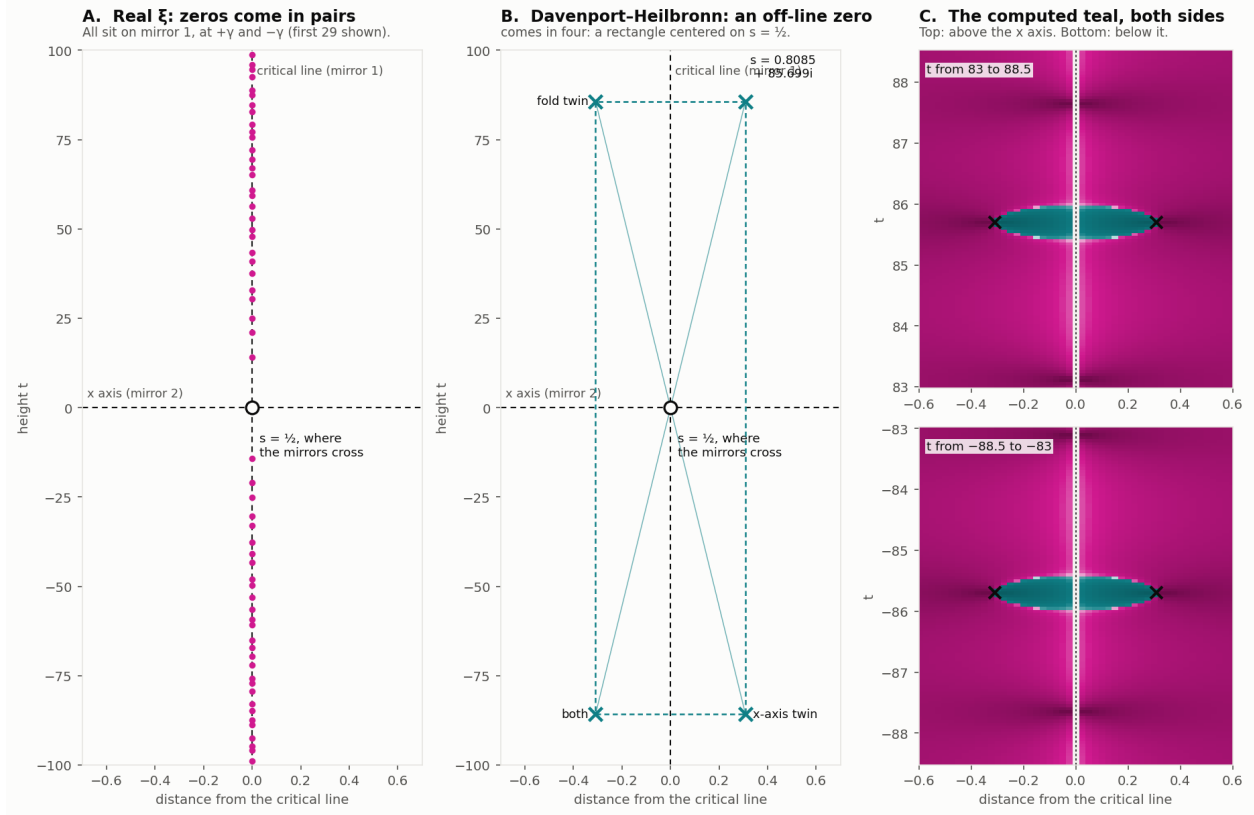


FIGURE 21. The second symmetry. Every zero at height t has a partner at $-t$. With both symmetries the picture is centered at $s = \frac{1}{2}$: on-line zeros come in pairs $\pm\gamma$, off-line zeros in quartets at the corners of a rectangle centered at $\frac{1}{2}$.

9. ALL ZEROS AT ONCE

Folding the plane across both symmetry lines sends each on-line pair $\pm\gamma$ to one point of an edge and each off-line quartet to a single interior point (Figure 22).

Computation 9.1. The four zeros of the first Davenport–Heilbronn quartet fold to the same point to 15 digits, at distance 52.9 from the edge in the coordinates of Figure 22; the first 1,000 zeros of ξ all fold onto the edge. Normalized nearest-neighbour gaps below 0.2 occur for 0.3% of the zeros, 0.8% of the eigenvalues of simulated GUE matrices and 18.1% of Poisson points, in line with Montgomery’s pair-correlation conjecture [29] and Odlyzko’s computations [31].

Computation 9.2. Place the first 1,000 zeros and their negatives on the diagonal of a $2,000 \times 2,000$ Hermitian matrix and conjugate by a random unitary matrix. Its eigenvalues return the zeros to 4×10^{-12} , and its characteristic function, completed by a smooth factor for the zeros above the 1,000th, reproduces Ξ on the line to 0.008% (Figure 23). The construction is circular: it is built from the zeros. The Hilbert–Pólya problem asks for a self-adjoint operator built from other data, such as the primes, whose spectrum is the set of zeros.

Under the map $w = 1 - 1/\rho$ the critical line goes to the unit circle, and Li’s criterion [25] states that RH holds if and only if $\lambda_n = \sum_{\rho} (1 - (1 - 1/\rho)^n) \geq 0$ for every $n \geq 1$.

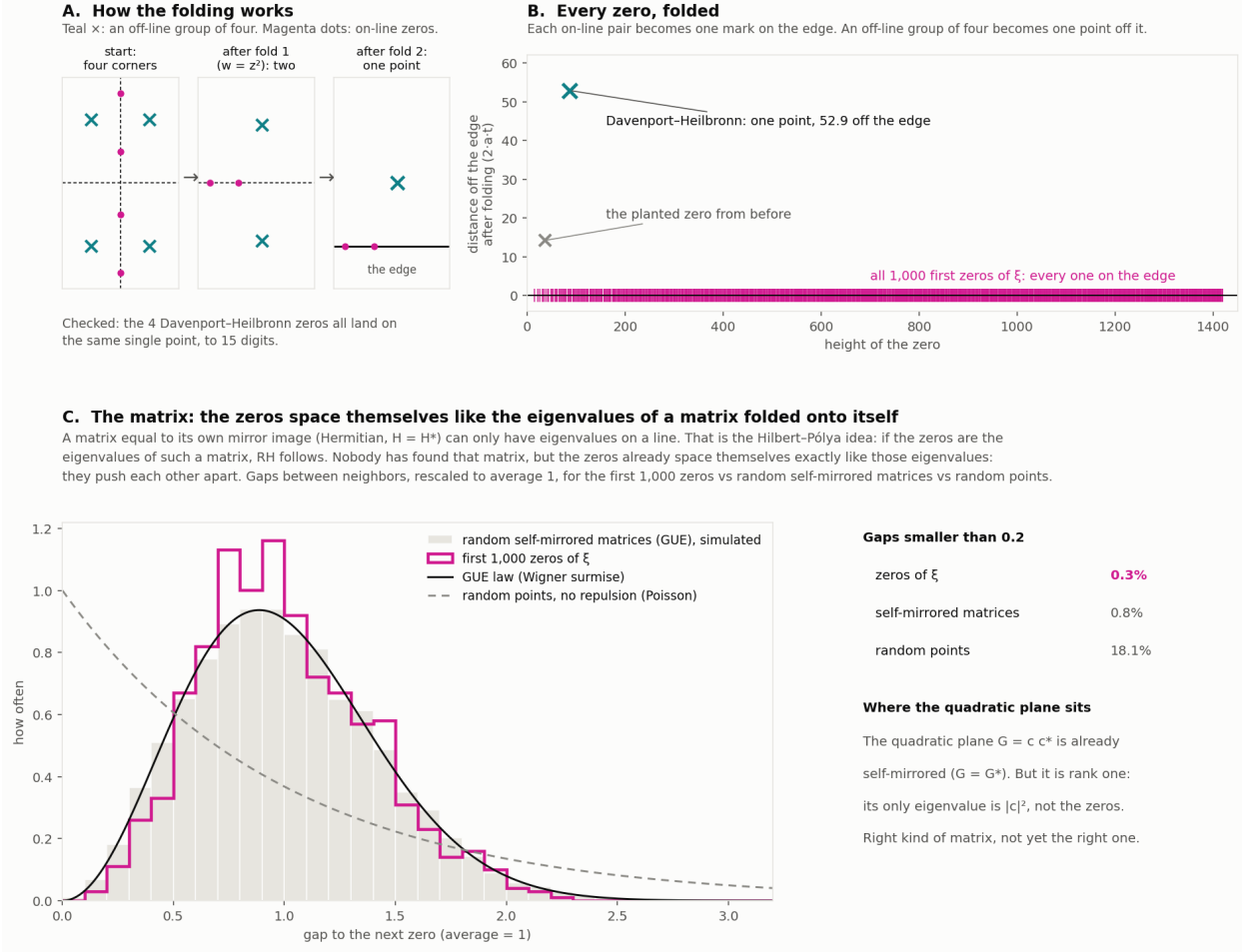


FIGURE 22. A: folding across both symmetry lines sends an off-line quartet to one point and an on-line pair to one point of the edge. B: all 1,000 first zeros of ξ sit on the edge; the Davenport–Heilbronn quartet sits off it. C: spacing statistics of the first 1,000 zeros against GUE and Poisson.

Proposition 9.3. *Summing over all nontrivial zeros, $\sum_{\rho} \frac{1}{\rho(1-\rho)} = 2 + \gamma - \log 4\pi = 0.0461914179\dots$, unconditionally. If RH holds, $\rho(1-\rho) = |\rho|^2$ for every zero.*

Proof. $1/(\rho(1-\rho)) = 1/\rho + 1/(1-\rho)$, and the symmetric sum $\sum_{\rho} 1/\rho = 1 + \frac{\gamma}{2} - \frac{1}{2} \log 4\pi$ follows from the Hadamard product of ξ [15, Ch. 12]. For $\rho = \frac{1}{2} + i\gamma_{\rho}$ with γ_{ρ} real, $\rho(1-\rho) = \frac{1}{4} + \gamma_{\rho}^2 = |\rho|^2$. $\square$

Computation 9.4. All 2,000 values $w = 1 - 1/\rho$ from the first 1,000 zero pairs lie on the unit circle to 2×10^{-16} , within 4.05° of $w = 1$; the first off-line Davenport–Heilbronn zero misses its circle by only 4×10^{-5} . The sum in Proposition 9.3 over 1,000 pairs plus a smooth tail agrees with $2 + \gamma - \log 4\pi$ to 3×10^{-7} .

9.1. The Taylor coefficients at $s = \frac{1}{2}$. The jet $c(0)$ of F at the center $s = \frac{1}{2}$ determines F , and hence every zero. By Proposition 8.1 its odd entries vanish, and its even entries are positive. RH can be phrased entirely in terms of this single jet: it is equivalent to the hyperbolicity of all Jensen polynomials built from it (Pólya), and Griffin, Ono, Rolin and Zagier proved hyperbolicity for each fixed degree and all sufficiently large shifts [19].

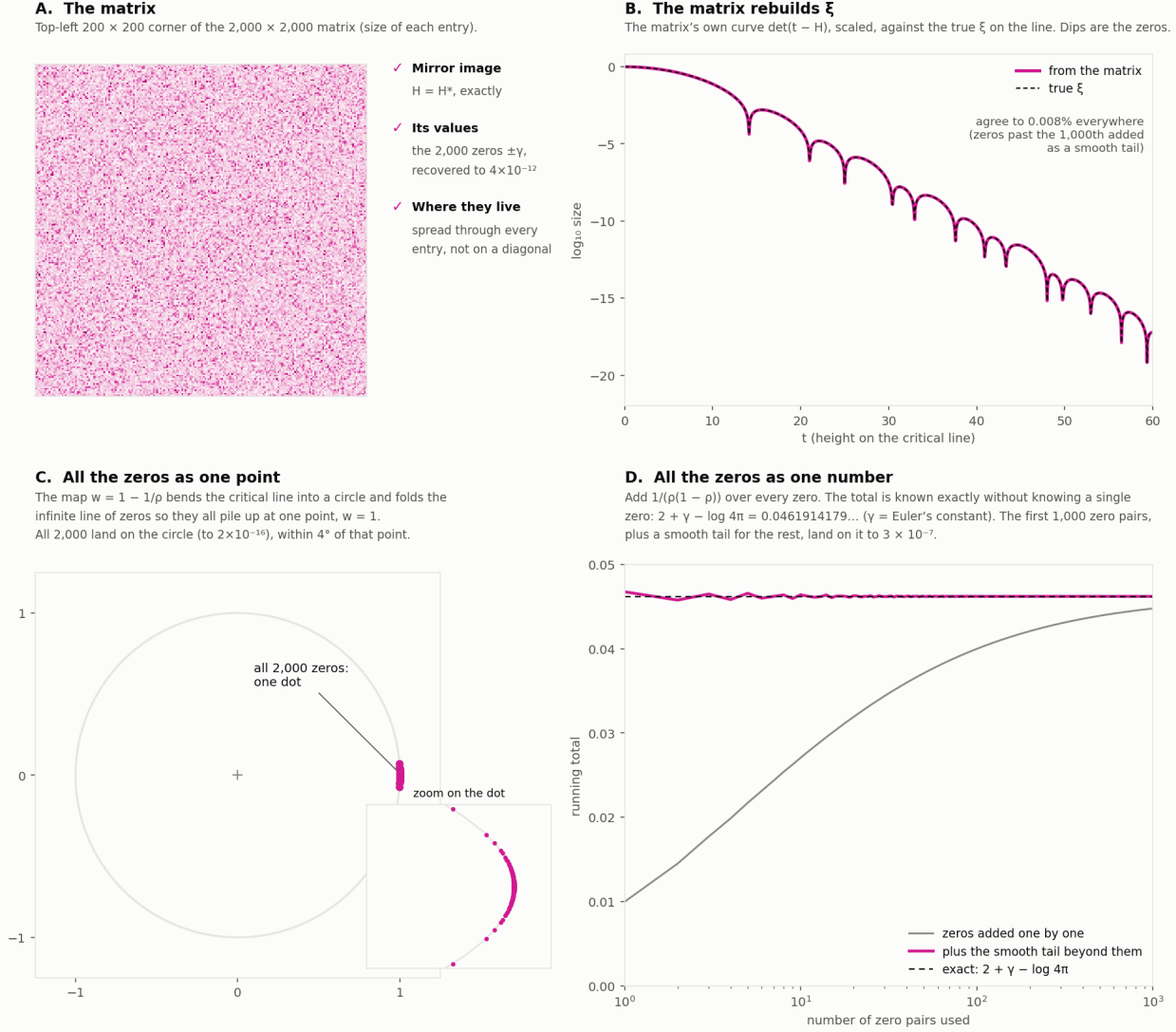


FIGURE 23. A: the $2,000 \times 2,000$ Hermitian matrix. B: its characteristic function against Ξ . C: the map $w = 1 - 1/\rho$ sends the zeros onto the unit circle, crowding them toward $w = 1$. D: partial sums of $1/(\rho(1 - \rho))$ approaching $2 + \gamma - \log 4\pi$.

Computation 9.5. At 110-digit precision the odd coordinates of $c(0)$ vanish to 88 digits. Truncating the Taylor series after the first N coordinates and locating the zeros of the truncation recovers all 14 zeros below height 62 to six decimals once $N = 260$ (Figure 24). All 279 Jensen polynomials tested (degrees 2 to 10, shifts 0 to 150) have only real roots; the same holds after planting an off-line zero at height 35.5, so low degrees do not detect a zero that high.

For a zero ρ the ratio $(1 - \bar{\rho})/\rho$ equals 1 if and only if $\text{Re } \rho = \frac{1}{2}$: every on-line zero maps to the single point 1.

Computation 9.6. All 1,000 first zeros of ξ , each re-found by Newton's method from a starting point with real part 0.62, land within 10^{-31} of the point 1. For the Davenport–Heilbronn function, every on-line zero up to height 200 lands on the point, and the four off-line quartets miss it by 0.0072, 0.0026, 0.0009 and 0.0025 (Figure 25).

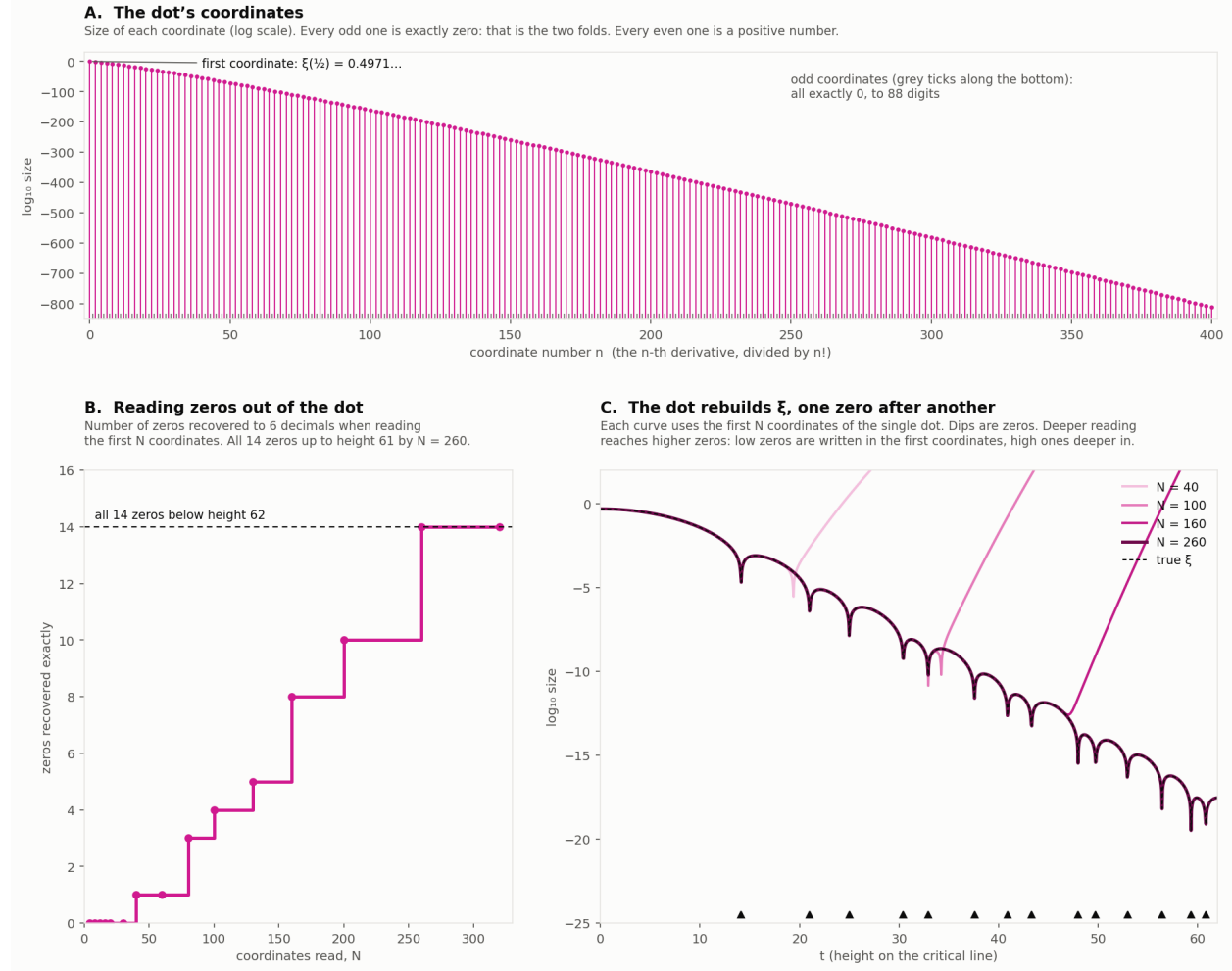


FIGURE 24. The Taylor jet of F at $s = \frac{1}{2}$. A: sizes of the coordinates; odd ones vanish. B: number of zeros recovered to six decimals from the first N coordinates. C: truncations rebuilding Ξ ; low zeros are carried by early coordinates, high zeros by later ones.

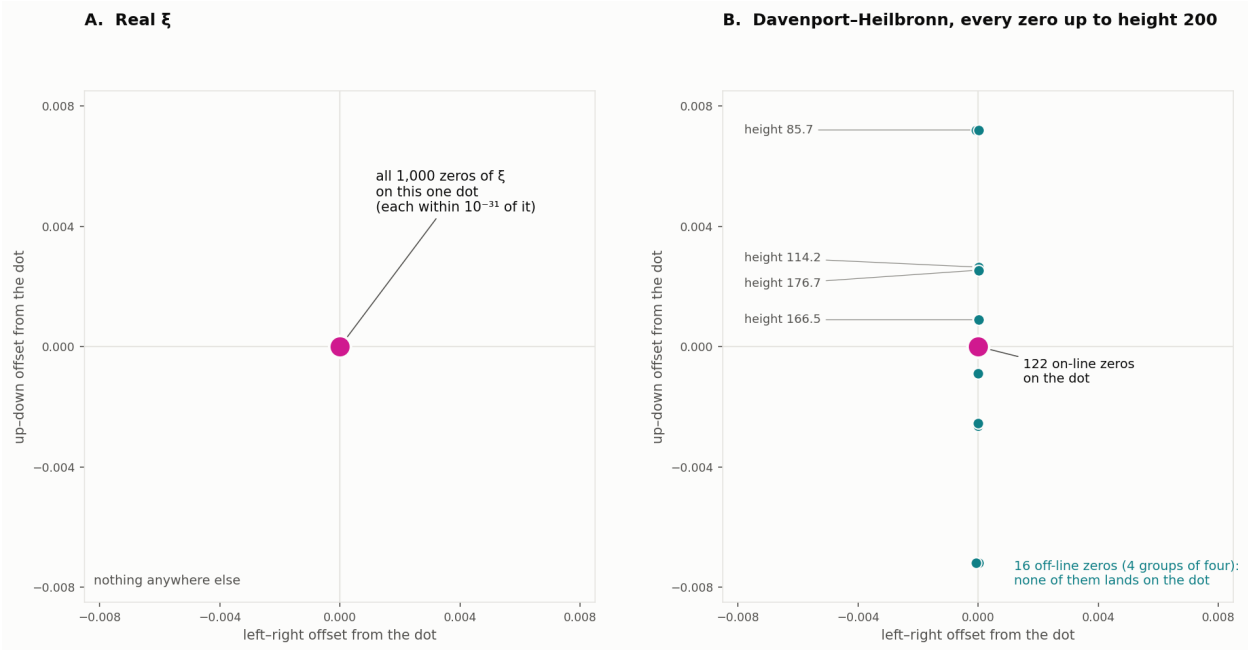


FIGURE 25. The ratio $(1 - \bar{\rho})/\rho$ for every zero. Left: the first 1,000 zeros of ξ , all at the single point 1. Right: the Davenport–Heilbronn zeros up to height 200; the on-line zeros sit at the point and the four off-line quartets do not.

10. THE COUNTING FUNCTION AS A STACK OF RINGS

Draw the n -th zero $\frac{1}{2} + i\gamma_n$ ($0 < \gamma_1 \leq \gamma_2 \leq \dots$) as a circle of circumference γ_n at height n . The resulting surface of revolution (a “horn”) has profile given by the inverse of the counting function.

Theorem 10.1 (Riemann–von Mangoldt; see [39]). $N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi e} + \frac{7}{8} + S(T) + O(1/T)$, where $S(T) = \pi^{-1} \arg \zeta(\frac{1}{2} + iT) = O(\log T)$.

Computation 10.2. For the first 1,000 zeros (circumferences 14.13 to 1,419.4) the growth in radius per ring averages 0.2239, against 0.2244 predicted by $1/\log r$, and the number of rings minus the smooth part of $N(T)$ stays within ± 0.73 . Only zeros on the line can be drawn as rings, while the counting law counts all zeros; for the Davenport–Heilbronn function the ring count therefore drops by exactly 2 at each off-line pair (heights 85.7, 114.2, 166.5, 176.7) and ends 8 short at height 200 (Figure 26).

A hypothetical off-line zero affects the three pictures used so far in different ways, summarized in Table 6.

Theorem 10.3 (von Koch [41]). *RH holds if and only if $\psi(x) = x + O(\sqrt{x} \log^2 x)$.*

Computation 10.4. Rebuild $(\psi(x) - x)/\sqrt{x}$ from the first 1,000 zeros and add one off-line quartet placed like the first Davenport–Heilbronn quartet ($\beta = 0.8085$, height 85.70). Its contribution grows like $x^{\beta-1/2}$: in units of $\sqrt{x}$ it reaches 1.9 at 10^6 , 16 at 10^9 and 110 at 10^{12} . The same planted quartet changes the Taylor coefficients of Computation 9.5: early coefficients by about 7%, 1.2% and 0.1% for a quartet at height 35.5, 85.7 and 300, and deep coefficients by more than 100% (Figure 27).

TABLE 6. Where a hypothetical off-line zero shows up.

Picture	Effect of an off-line pair at height γ_0
Ring count	a deficit of 2 from height γ_0 onward; rings below γ_0 are unaffected
Taylor coefficients at $s = \frac{1}{2}$	every coefficient moves, the early ones by an amount that decreases with γ_0
Prime counting	a wave through all x that outgrows $\sqrt{x}$ (Theorem 10.3)

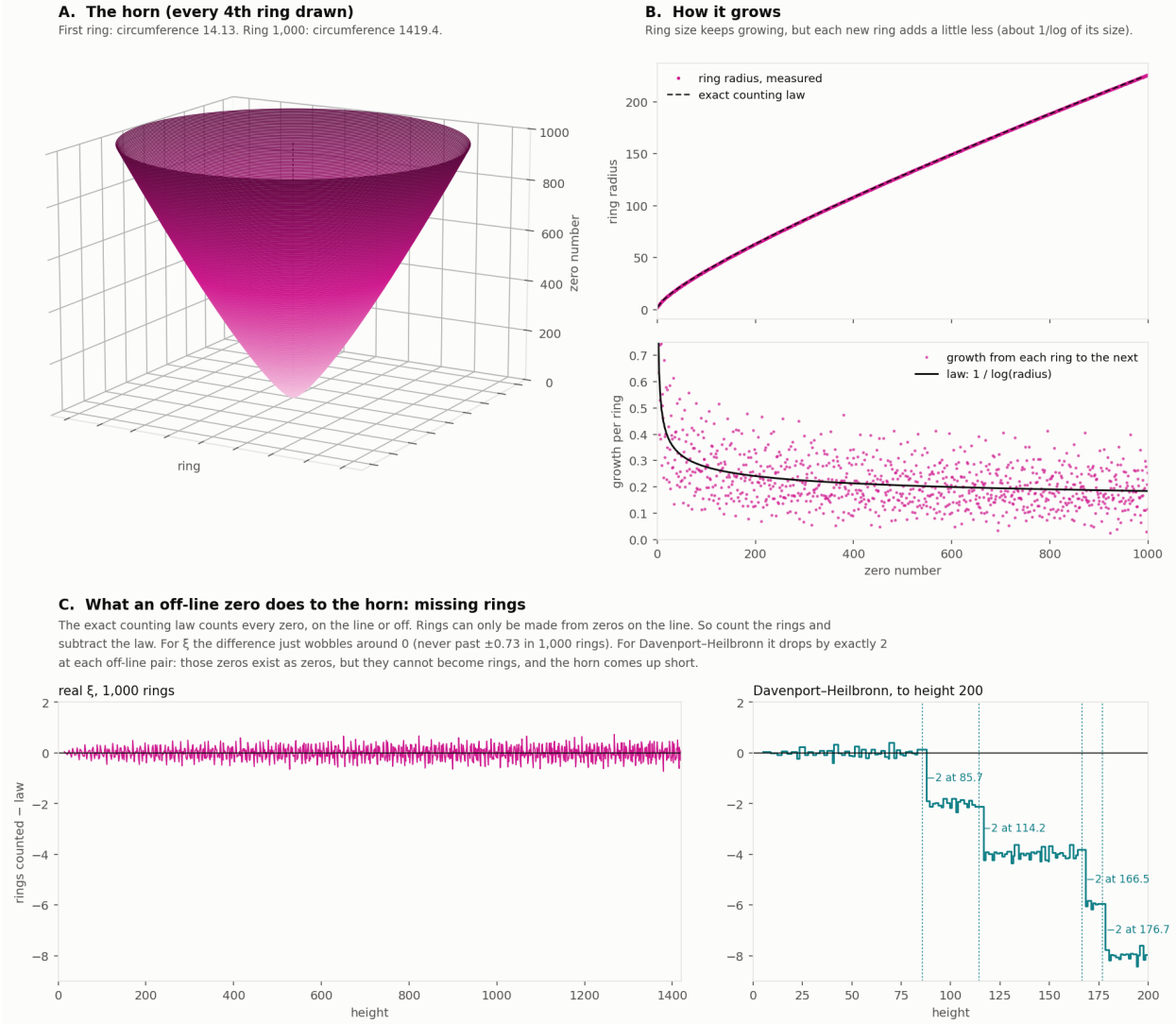


FIGURE 26. A: the first 1,000 zeros as rings (every fourth ring drawn). B: ring radius against the counting law, and the growth per ring against $1/\log r$. C: rings counted minus the counting law, for ξ (left) and for the Davenport-Heilbronn function (right), whose count drops by 2 at each off-line pair.

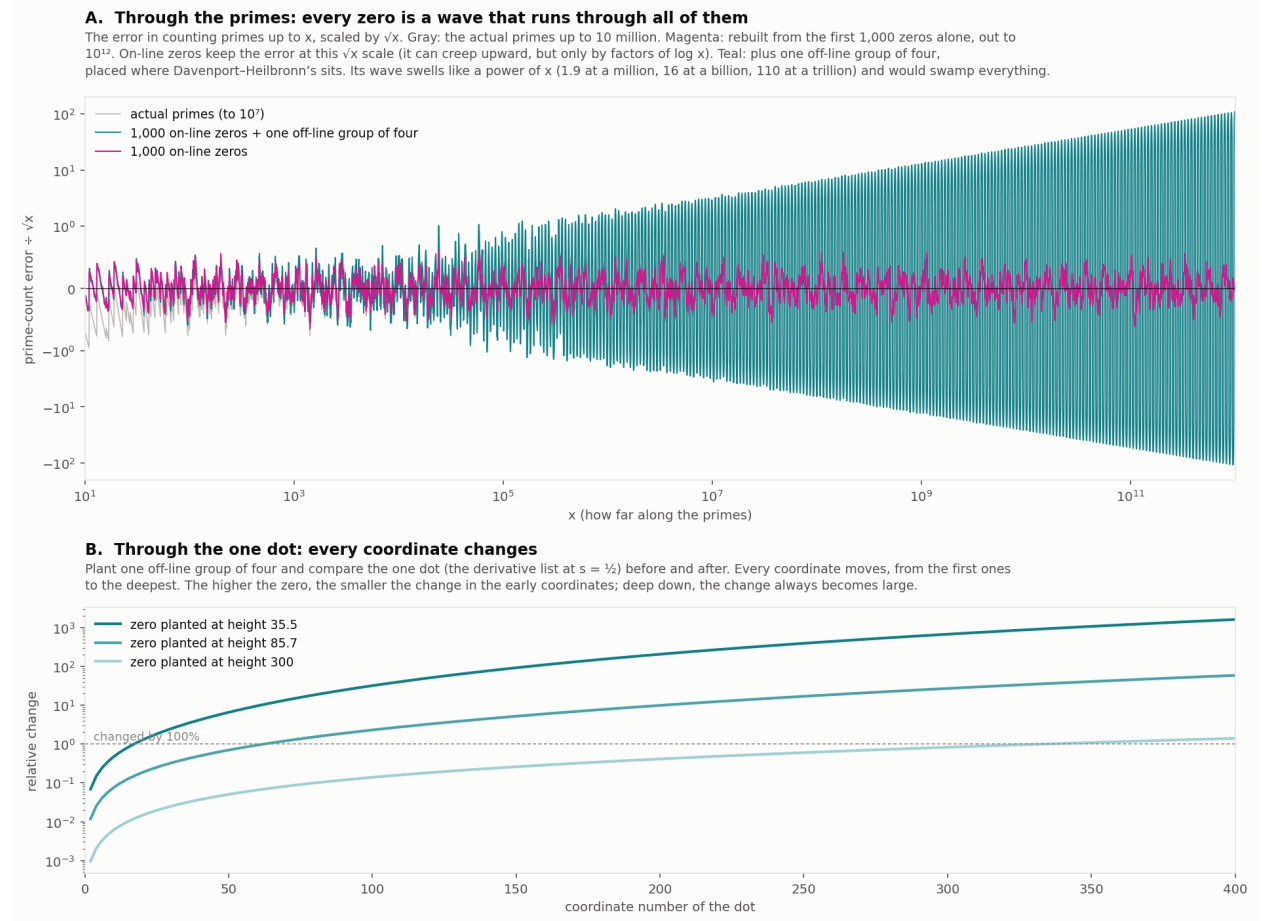


FIGURE 27. A: $(\psi(x) - x)/\sqrt{x}$ for the actual primes to 10^7 (gray), rebuilt from the first 1,000 zeros to 10^{12} (magenta), and with one off-line quartet added (teal). B: relative change of the Taylor coefficients at $s = \frac{1}{2}$ caused by planting one off-line quartet at three heights.

11. RIGIDITY OF THE ZEROS AND THE PROPORTION RESULTS

Computation 11.1. Using all 22,491 zeros up to height 20,000 (the count above height 100 is 22,462, against 22,462.4 from the counting law), the number variance of the ring count in windows of L mean spacings saturates near 0.3 to 0.4, while for GUE it continues to grow (0.69 at $L = 30$) (Figure 28A). Berry’s semiclassical formula [4], in which the primes play the role of short periodic orbits, predicts this saturation.

Table 7 collects what is proved about the zeros from the arithmetic side.

TABLE 7. Proved facts about the zeros.

Statement	Source
$N(T)$ obeys the Riemann–von Mangoldt formula	[39]
No zeros with $\sigma \geq 1 - 1/(5.558691 \log t)$, $t \geq 2$	[30]
All zeros with $0 < \gamma \leq 3 \times 10^{12}$ lie on the line	[33]
At least 41.72% of zeros lie on the line	[34]
More than two thirds of zeros are simple and on the line	[1]
All zeros lie on the line (RH)	open

Remark 11.2 (The two-thirds argument, schematically). As we read [1], Weil’s Hermitian form is compressed to a finite matrix by testing it against a family of identical windows over heights $[T, 2T]$. In this compression each simple zero on the line contributes a positive direction of unit trace, while an off-line pair contributes a block of signature $(1, 1)$, which Sylvester’s law of inertia detects. The trace counts zeros, while the squared Frobenius norm is computed from the primes through second moments of $\Lambda(n)$ and equals $4/3$ per zero for the simplest window; a rank–trace inequality then forces at least $2 - 4/3 = 2/3$ of the zeros onto the line. The ingredients (finite Weil matrices and inertia counts) are the same as those used in Section 2. A positive proportion, even 100%, does not imply RH. We refer to [1] for the actual argument.

Computation 11.3. The odd floor falls super-exponentially: $\lambda(0.5) = 1.94 \times 10^{-4}$, $\lambda(0.6) = 5.97 \times 10^{-7}$, $\lambda(0.8) = 1.57 \times 10^{-14}$, $\lambda(1.0) = 1.49 \times 10^{-26}$, and Zhu’s reference value at $a = 1.2$ is 4.76×10^{-45} [46]. Positivity is certified for $a \leq 0.8$ in [46]; the unpublished checkpoint CP22 of [28] reports a certified lower bound on $0 < a \leq \frac{3}{5}$ (Figure 28D).

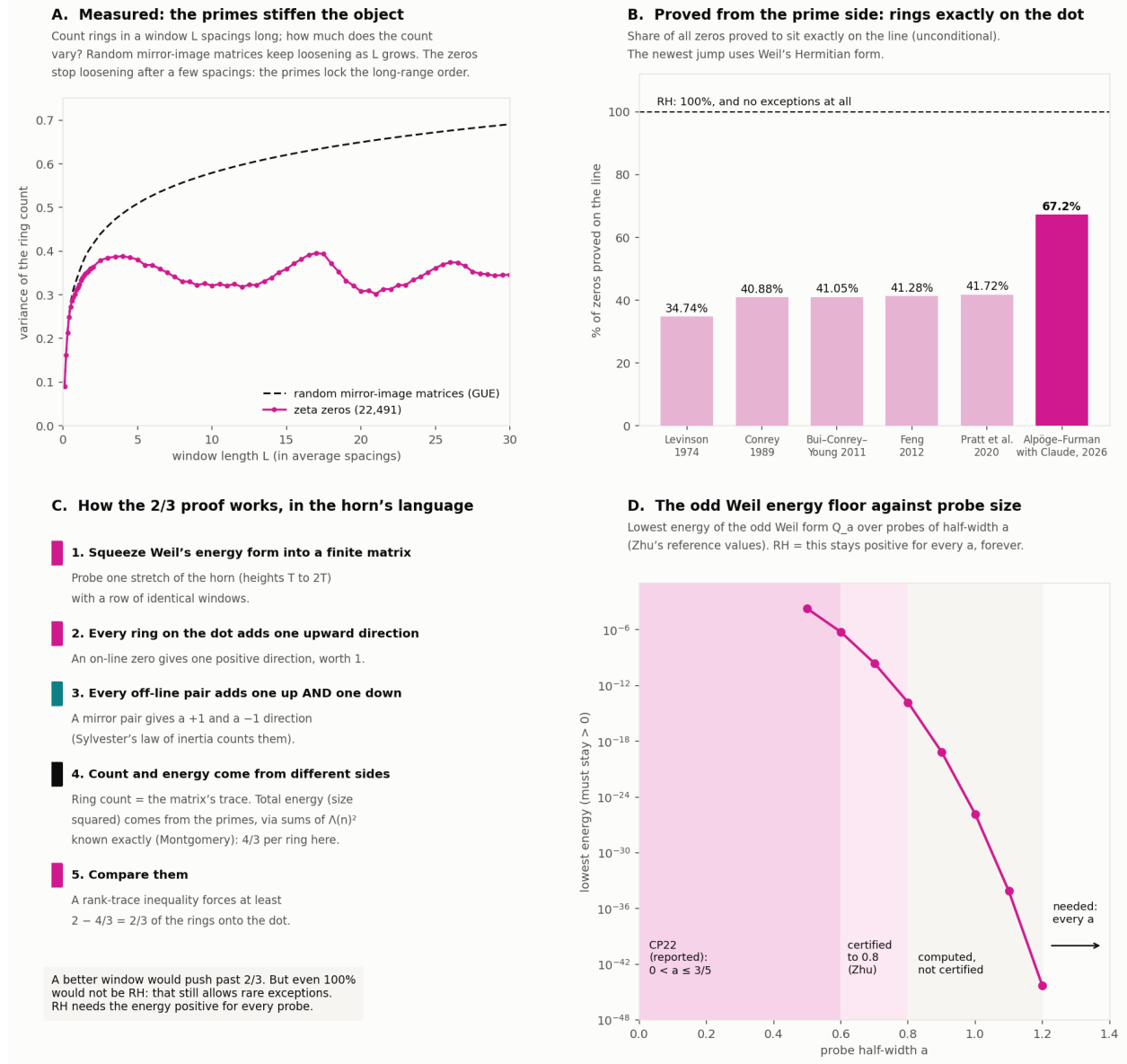


FIGURE 28. A: number variance of the zeros (Computation 11.1) against GUE. B: proved lower bounds for the proportion of zeros on the line: Levinson [24], Conrey [11], Bui-Conrey-Young [7], Feng [18], Pratt-Robles-Zaharescu-Zeindler [34] and Alpöge-Furman [1]. C: the two-thirds argument, schematically. D: the odd Weil floor against probe half-width, with the certified ranges.

12. A LIGHT-CONE READING OF THE ODD WEIL FORM

Proposition 12.1 (Finite propagation). *If $\text{supp } f \subseteq [-a, a]$ then $\text{supp}(f * \tilde{f}) \subseteq [-2a, 2a]$, so Q_a involves only the prime powers with $\log n \leq 2a$. The set of such n grows with a at unit speed in $\log n$ per half unit of a .*

Proposition 12.2 (Monotonicity). *$\lambda(a)$ is nonincreasing in a .*

Proof. A test function supported in $[-a, a]$ is supported in $[-a', a']$ for every $a' > a$, and $Q_{a'}$ agrees with Q_a on it. $\square$

Proposition 12.3. *Let f be real and odd with support in $[-a, a]$, and put $F_f(t) = \hat{f}(t)/i$, a real odd entire function of exponential type a . If RH holds then $Q_a(f) = 2 \sum_{\gamma > 0} F_f(\gamma)^2$. In general, an off-line quartet with $\rho = \frac{1}{2} + \delta + i\gamma_0$ contributes $4 \operatorname{Re} F_f(\gamma_0 + i\delta)^2$, which can be negative.*

Proof. For odd real f , $\hat{g}(t) = \hat{f}(t)\hat{f}(-t) = -\hat{f}(t)^2 = F_f(t)^2$, analytically in t . Insert this into (1). On the line the terms at $\pm\gamma$ are equal and nonnegative. For a quartet the four arguments are $\pm\gamma_0 \pm i\delta$; oddness and reality give $F_f(-w)^2 = F_f(w)^2$ and $F_f(\bar{w}) = \overline{F_f(w)}$, so the four terms add to $4 \operatorname{Re} F_f(\gamma_0 + i\delta)^2$. $\square$

Heuristic 12.4 (A horizon on the zero side). By the Cartwright–Levinson theorem [23] the zeros of F_f on the positive axis have density a/π , while the zeta zeros have density $\frac{1}{2\pi} \log \frac{t}{2\pi}$ at height t . The two densities agree at

$$T^*(a) = 2\pi e^{2a}.$$

Below $T^*(a)$ a minimizer can afford a zero of F_f at every zeta zero; above it the zeta zeros are too dense, and F_f must instead be small. One therefore expects that (i) the minimizer’s zeros track the zeta zeros below $T^*(a)$, (ii) the energy $\int F_f^2$ is concentrated below $T^*(a)$, (iii) $\lambda(a)$ is paid almost entirely by zeros above $T^*(a)$, and (iv) $-\log \lambda(a)$ grows roughly in proportion to $T^*(a)$, in line with the Landau–Widom law identified in [46].

Computation 12.5 (The light-cone test). For $a \in \{0.4, 0.6, 0.8, 1.0, 1.1\}$ we computed the minimizer f with 72 modes at 640 bits and evaluated $F_f(t) = 2\sqrt{a} \sum_i c_i N_i(-1)^i j_{2i+1}(at)$ exactly, using spherical Bessel functions for $t < 150$ and the exact finite endpoint expansion of $\hat{f}$ above (the two agree to 10^{-99} at $t = 150$). We summed $2F_f(\gamma)^2$ over all 22,491 zeros up to height 20,000 (the first 80 at 50 digits) and estimated the tail from the endpoint value $f(a)$. Table 8 and Figures 29 and 30 show the results (the last column of the table is the distance from γ_1 to the nearest zero of F_f):

- (1) The matrix floor $\lambda(a)$ and the zero sum agree to within 2×10^{-5} at every a tested, an end-to-end check of (1) and of the normalization of the engine.
- (2) The fraction of $\int_0^\infty F_f^2$ lying above $T^*(a)$ is 7×10^{-6} at $a = 0.6$ and 8×10^{-24} at $a = 1.1$.
- (3) Zeros below $T^*(a)$ contribute at most 10^{-5} of $\lambda(a)$. The minimizer places a zero of F_f within 2.6×10^{-32} of γ_1 at $a = 1.1$; its zeros stay within 0.01 of the zeta zeros up to 1.2 to 1.7 horizons.
- (4) On $0.8 \leq a \leq 1.1$, $-\log \lambda(a)$ is fitted by $1.83 T^*(a) - 25.4$; extrapolated to $a = 1.2$ the fit gives about 1×10^{-44} , against Zhu’s independent value 4.76×10^{-45} . The local rate $d(-\log \lambda)/dT^*$ still increases slowly (1.65 at $a = 0.6$, 1.91 at $a = 1.08$), so this is a fit, not a law.

TABLE 8. The light-cone test (Computation 12.5).

a	$T^*(a)$	$\lambda(a)$	zero sum/ λ	λ -share below T^*	energy above T^*	miss at γ_1
0.4	14.0	1.47×10^{-2}	0.999993	0	2.2×10^{-2}	0.3
0.6	20.9	5.96×10^{-7}	0.999983	9.6×10^{-6}	7.1×10^{-6}	4.9×10^{-5}
0.8	31.1	1.56×10^{-14}	0.999995	1.0×10^{-7}	8.1×10^{-11}	2.8×10^{-12}
1.0	46.4	1.49×10^{-26}	0.999999	1.6×10^{-13}	3.5×10^{-18}	4.3×10^{-24}
1.1	56.7	7.68×10^{-35}	0.999983	8.2×10^{-16}	8.1×10^{-24}	2.6×10^{-32}

Remark 12.6. That the minimizer’s Fourier transform has zeros extremely close to the zeta zeros is known: Connes, Consani and Moscovici observe it for the even minimizer with the primes up to 13, with the first zero reproduced to 2.5×10^{-55} [10], and Groskin reaches about 300 digits for the first ten zeros with a related truncated form [20]. Computation 12.5 adds the explicit-formula balance, the split of $\lambda(a)$ across the horizon, and the dependence of the tracking range on a .

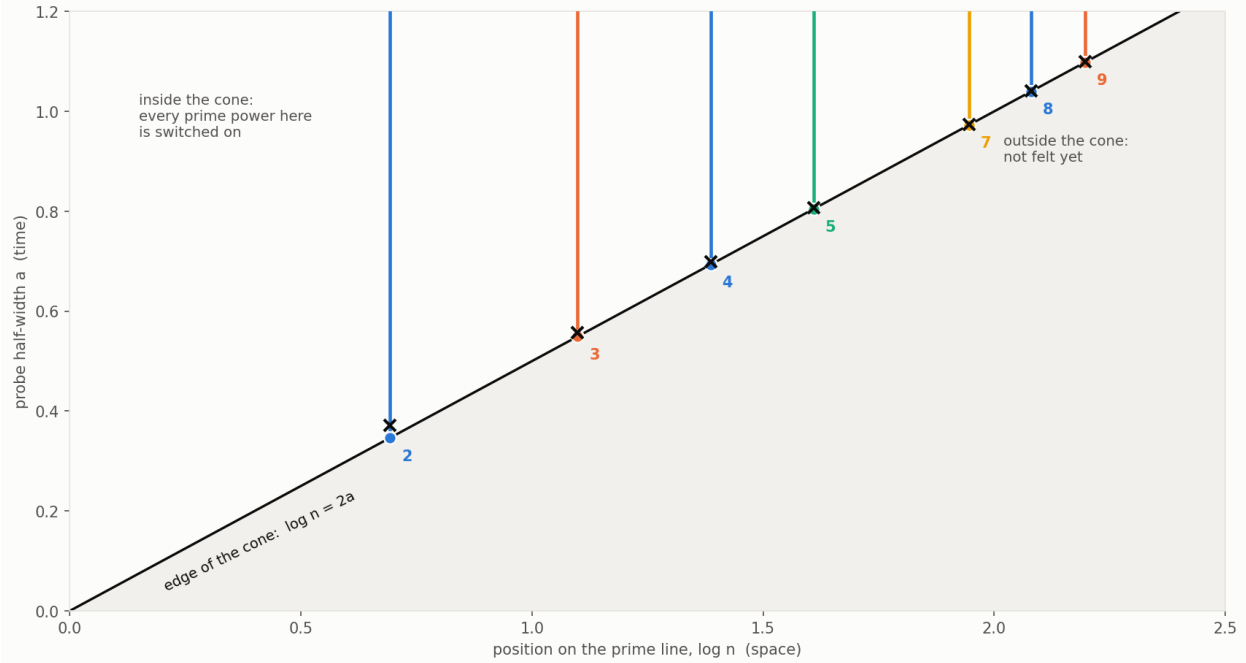


FIGURE 29. Proposition 12.1 drawn as a light cone. Vertical: probe half-width a . Horizontal: $\log n$. Dots mark each prime power's entry at $a = \frac{1}{2} \log n$; crosses mark where the floor fails if that newest prime power is deleted (Table 1).

Remark 12.7 (The physical analogy). Proposition 12.1 is a finite-propagation statement, and Proposition 12.2 says the “energy” $\lambda(a)$ never increases as the cone opens. In physics finite speed together with a conserved or bounded-below energy gives stability; here the monotonicity points the other way, and nothing prevents $\lambda(a)$ from approaching 0. Computation 12.5 also shows where a lower bound would have to come from: the primes inside the cone determine Q_a exactly, but the size of $\lambda(a)$ is set by the zeta zeros beyond $T^*(a)$. By Proposition 12.3, a hypothetical off-line zero would become visible once the horizon passes its height.

Question 12.8. Is there a constant C such that $\lambda(a) \geq \exp(-C \cdot 2\pi e^{2a})$ for all $a > 0$? An affirmative answer gives $\lambda(a) > 0$ for all a , hence RH through the odd-sector criterion. For $a \leq 1.2$ the data are consistent with $C = 1.5$ (the largest observed value of $-\log \lambda(a)/T^*(a)$ is 1.47, at $a = 1.2$).

Remark 12.9. For the Davenport–Heilbronn function the zero density is $\frac{1}{2\pi} \log \frac{5t}{2\pi}$ (conductor 5), so the analogous horizon is $\frac{2\pi}{5} e^{2a}$, which reaches the first off-line zero (height 85.7) near $a \approx 2.1$. The heuristic predicts that the corresponding odd floor becomes negative near that value. We have not tested this.

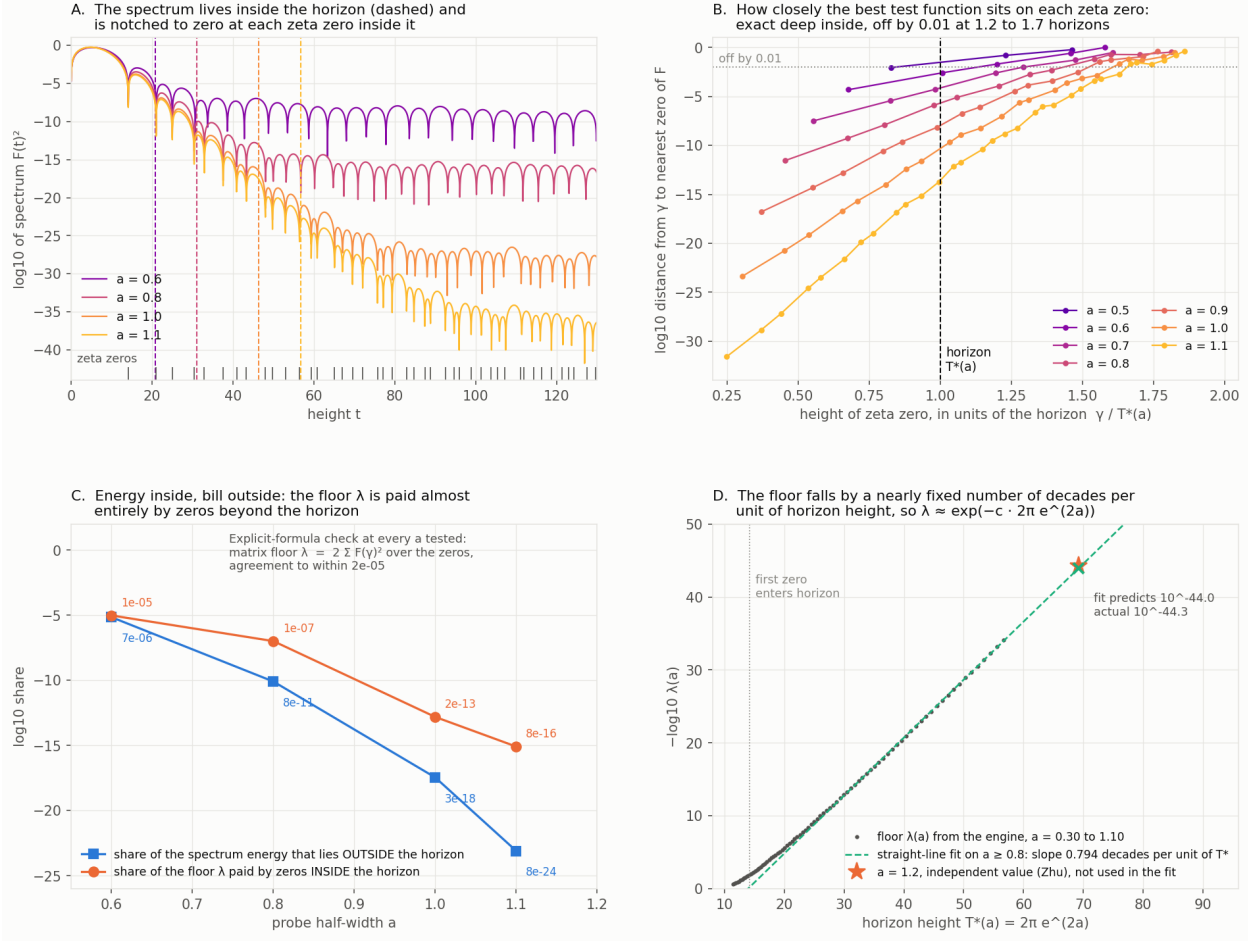


FIGURE 30. A: $F_f(t)^2$ for the minimizer at four widths; dashed lines are the horizons $T^*(a)$, ticks are zeta zeros. B: distance from each zeta zero to the nearest zero of F_f , with height measured in units of $T^*(a)$. C: share of the energy of F_f above $T^*(a)$ and share of $\lambda(a)$ contributed by zeros below $T^*(a)$. D: $-\log_{10} \lambda(a)$ against $T^*(a)$ with a linear fit on $a \geq 0.8$ and Zhu's value at $a = 1.2$ (not used in the fit).

13. SCALING, COMOVING NORMALIZATION AND A MEAN-SQUARE CONSTANT

By the explicit formula for ψ [15, Ch. 17],

$$(3) \quad \psi_0(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \log 2\pi - \frac{1}{2} \log(1 - x^{-2}),$$

where ψ_0 is ψ normalized at its jumps and the sum over zeros is taken symmetrically.

Proposition 13.1 (Scaling). *The term of a zero $\rho = \frac{1}{2} + i\gamma$ in (3) is $-\sqrt{x} e^{i\gamma \log x} / \rho$. Its local wavelength in x is $2\pi x / |\gamma|$, so between any two points $x_1 < x_2$ the wavelengths of all zero terms scale by the same factor x_2/x_1 . In the variable $u = \log x$ each term oscillates with the fixed frequency γ , and after division by $\sqrt{x}$ its modulus is the constant $1/|\rho|$. A term with $\text{Re } \rho = \beta$ has modulus $x^{\beta-1/2}/|\rho|$ after the same normalization.*

Proof. The phase $\gamma \log x$ has derivative γ/x in x . □

In the language of cosmology, x plays the role of the scale factor, $u = \log x$ is a comoving coordinate, the measure dx/x is invariant under $x \mapsto cx$, and the uniform stretch of every wavelength by x_2/x_1

is the analogue of an achromatic redshift. The normalization by $\sqrt{x}$ is the unique power that makes the dilation $f(x) \mapsto c^{1/2}f(cx)$ unitary on $L^2(\mathbb{R}_+, dx)$; the same $\frac{1}{2}$ appears in the Berry–Keating operator $\frac{1}{2}(xp + px)$ [5].

Theorem 13.2 (Cramér [13]). *Assume RH. Then*

$$\frac{1}{\log X} \int_2^X \frac{(\psi(x) - x)^2}{x^2} dx \longrightarrow \sum_{\rho} \frac{1}{|\rho|^2} = 2 + \gamma - \log 4\pi = 0.0461914\dots$$

The integral is the mean of $((\psi(x) - x)/\sqrt{x})^2$ with respect to $du = dx/x$, and the value of the limit follows from Proposition 9.3. Without RH the same mean would grow like a positive power of X .

Computation 13.3. With every prime power up to 10^{10} (exact segmented sieve, ψ sampled at 32,840 points with spacing 5×10^{-4} in $\log x$), let $S(x) = (\psi(x) - x + \log 2\pi + \frac{1}{2} \log(1 - x^{-2}))/\sqrt{x}$ (Figure 31).

- (1) The running mean of S^2 over $\log x$ from 10^2 to 10^{10} is 0.0458, within 0.9% of $2 + \gamma - \log 4\pi$; its value over each single decade lies between 0.042 and 0.048.
- (2) In two windows about 1,800 times apart in x ($3 \times 10^3 \leq x \leq 3 \times 10^6$ and $3 \times 10^6 \leq x \leq 10^{10}$) the spectrum of S in u peaks within 0.04 of each of the first six zeta zeros.
- (3) The explicit formula with the first 1,000 zeros reproduces S with correlation 0.983.
- (4) Adding one off-line quartet placed like the first Davenport–Heilbronn quartet makes the same running mean exceed 35 at 10^{10} , still increasing.

Remark 13.4. The same normalization applied to Section 12 gives $T^*(a)/e^{2a} = 2\pi$ for every a , and the ratio $-\log \lambda(a)/T^*(a)$ equals 1.39 at $a = 1.1$ and 1.47 at $a = 1.2$. Whether this ratio converges is a reformulation of the rate question in Computation 12.5(4), and a bound on it is exactly Question 12.8.

Remark 13.5 (Heuristic dictionary). We record some analogies that motivated the pictures. They make no claim about physical cosmology.

- (1) A medium whose refractive index is constant in time slows a wave but preserves its frequency; a spatially uniform medium whose index increases in time lowers the frequency (“time refraction”), and wave propagation in an expanding homogeneous universe can be written in this form. In Proposition 13.1 the frequencies are preserved in the comoving variable u and lowered only when measured in x .
- (2) On the modular surface $\mathrm{SL}_2(\mathbb{Z}) \backslash \mathbb{H}$ the scattering matrix is $\varphi(s) = \xi^*(2s - 1)/\xi^*(2s)$ with $\xi^*(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$ [17, 22]. A zero ρ of ζ gives a resonance at $s = \rho/2$, which lies on $\mathrm{Re} s = \frac{1}{4}$ exactly when $\mathrm{Re} \rho = \frac{1}{2}$; so RH says that all these resonances have the same decay rate. Burnol has recast RH in a Lax–Phillips framework as a causality property (see the survey [42]).
- (3) In Connes’ trace formula the critical zeros appear as an absorption spectrum, and noncritical zeros would appear as resonances [9, 42]. Sierra constructs a massless Dirac fermion in Rindler spacetime with reflecting boundaries whose periods, measured by the accelerated observer, are $\log p$, and obtains the Riemann zeros as energy levels in a limit [37].

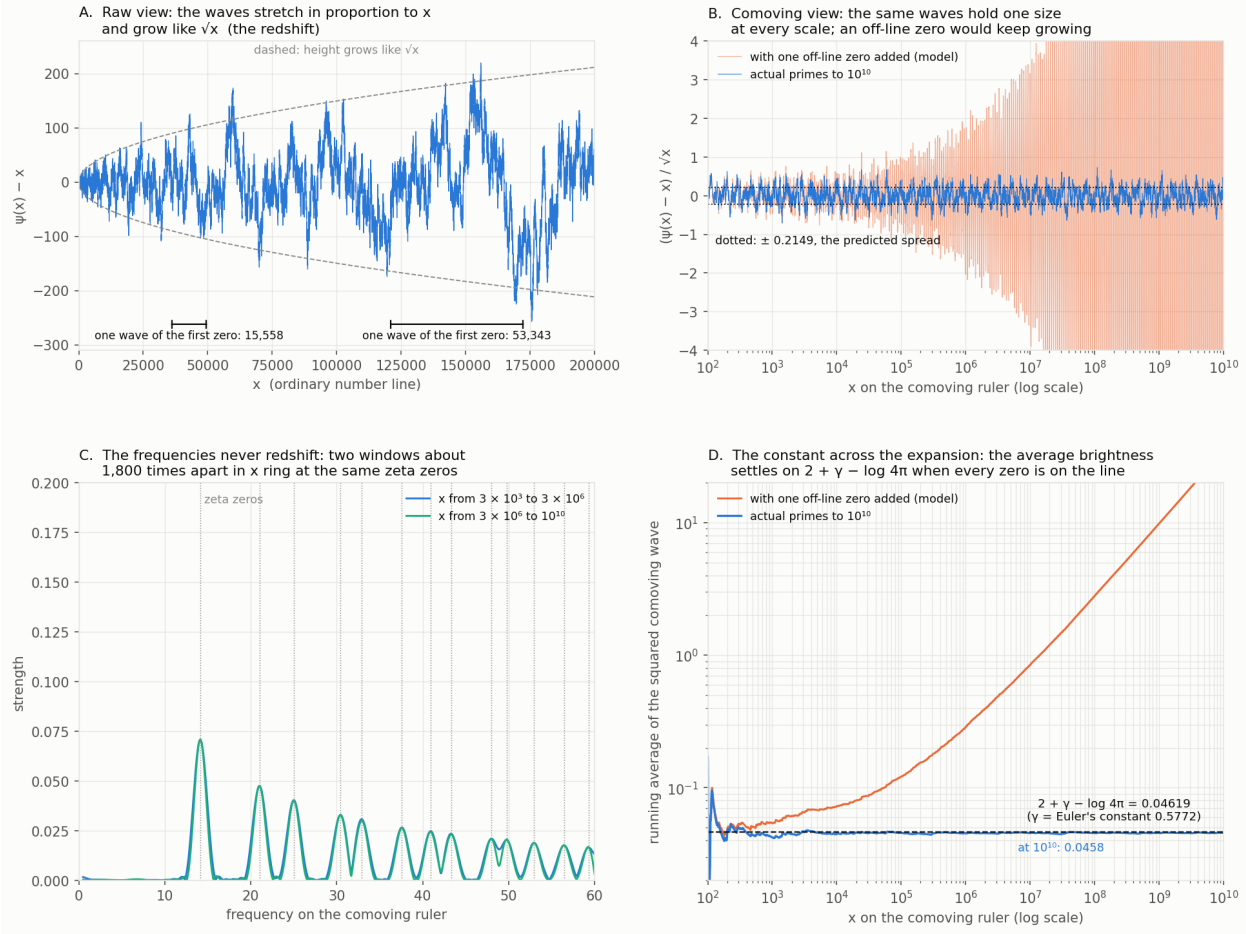


FIGURE 31. A: $\psi(x) - x$ on a linear scale up to 2×10^5 ; the wavelength of the first zero's term is 15,558 near $x = 35,000$ and 53,343 near $x = 120,000$, and the amplitude grows like $\sqrt{x}$. B: the same signal divided by $\sqrt{x}$ against $\log x$ from 10^2 to 10^{10} ; dotted lines mark ± 0.2149 , the square root of the constant in Theorem 13.2. Orange: with one off-line quartet added. C: spectra of the normalized signal in two windows about 1,800 times apart in x . D: running mean of the squared normalized signal against $2 + \gamma - \log 4\pi$.

14. OPEN QUESTIONS

- (1) Find a prime-side mechanism, uniform in a , explaining why the relay of Computation 2.1 never fails; equivalently, prove $\lambda(a) \geq 0$ for all a .
- (2) Question 12.8: prove a lower bound $\lambda(a) \geq \exp(-C \cdot 2\pi e^{2a})$, and determine whether $-\log \lambda(a)/T^*(a)$ converges.
- (3) Test the horizon heuristic on the Davenport–Heilbronn control, where it predicts a negative floor near $a \approx 2.1$.
- (4) Extend Computation 13.3 to 10^{12} ; the cross terms between zeros in the running mean decay only like $1/\log x$.
- (5) Rebuild the two-thirds argument of [1] inside the finite odd Weil matrices of Section 2, and search for windows with a prime-side constant below $4/3$.

APPENDIX A. METHODS AND REPRODUCIBILITY

Table 9 lists the computational engines, their working precision, and the checks each passed. Weil-form computations used python-flint (Arb ball arithmetic, with midpoints used for the linear algebra), zeta computations used mpmath, and the sieves were written in C.

TABLE 9. Engines and validation.

Engine	Computes	Precision	Validation
Odd Weil matrix (double)	Q_a in 24 odd Legendre modes	53 bits	reproduces Zhu's floors for $0.5 \leq a \leq 0.8$
Odd Weil matrix (high precision)	exact rational kernel polynomials, inertia counts, inverse iteration	420 bits (40 modes), 640 bits (72 modes)	floors at $a = 0.8, 0.9, 1.0$ match Zhu to three significant figures
Segmented sieve (C, two threads)	every prime gap to 10^{11} : histograms, records, residue transitions	exact	$\pi(10^9)$, $\pi(10^{11})$ exact; all 40 record gaps below 10^{11} match [32, 44]
ψ sampler (C)	$\psi(x)$ at 32,840 log-spaced points to 10^{10}	exact integers, compensated sums	matches a direct count to 8×10^{-9} at 2,018 points below 10^6
Riemann–Siegel root finder	all 22,491 zeros up to height 20,000	about 10^{-3}	count above height 100 is 22,462 against 22,462.4
High-precision ξ (mpmath)	derivative jets by Cauchy integrals, growth maps, Taylor coefficients at $\frac{1}{2}$	15 to 110 digits	$\xi(\frac{1}{2}) = 0.49712077818831 \dots$; odd coefficients vanish to 88 digits
Davenport– Heilbronn	Hurwitz-zeta representation; line scan, grid search and Newton	15 to 20 digits	reproduces the zero at $0.80851718245 + 85.69934848538i$; 130 zeros to height 200 against 129.5
Hermitian test matrix	2000×2000 matrix from the first 1,000 zero pairs	53 bits	eigenvalues returned to 4×10^{-12}
Light-cone test	minimizer spectrum by spherical Bessel sums and exact endpoint expansion	640 bits	two formulas agree to 10^{-99} ; zero sum matches λ to 2×10^{-5}

Limits. All values of $\lambda(a)$ outside the ranges certified in [46] are numerical, not certified; interval radii were not propagated through the linear algebra. At $a = 0.6$ the 72-mode floor (5.956×10^{-7}) lies 0.3% below the 40-mode value, as expected, since enlarging the trial space can only lower a minimum. Zero statistics use Riemann–Siegel accuracy (about 10^{-3}), adequate for spacing statistics but not for certification. Berry's saturation, Montgomery's pair correlation and the Hardy–Littlewood weights are tested here, not proved. Scripts and data are available from the author on request.

APPENDIX B. SUMMARY OF STATEMENTS

#	Statement	Status	Where
1	Each prime power $2, \dots, 9$ is load-bearing at its entry	Computation	2.1
2	Color 2 pays 97.6–98.8% of the deficit at $a = 0.8$ – 1.0	Computation	§2
3	Record alignments of d colors improve like $k^{-1/d}$	Computation; Dirichlet	3.2
4	Prime line minus prime-power waves equals the zero bumps	Explicit formula; computed to 5×10^{-14}	4.2

#	Statement	Status	Where
5	Prime-power weights lie on $\frac{1}{k} 2xe^{-x/2}$	Proposition	4.1
6	Chords land on products; primes are never hit	Proposition	5.1
7	Upper edge is the staircase $\max\{p, (x-2q)/(q-2)\}$	Proposition	5.3
8	A prime q has $q-2$ reach-backs with slopes $1/(q-b)$	Proposition	5.6
9	Teeth converge to one triangle after scaling	Proposition	6.1
10	Residues of gaps alternate after removing multiples of 6	Proposition; checked to 10^{11}	6.2, 6.3
11	Gap frequencies follow the Hardy–Littlewood weights	Conjecture; median deviation 2.2%	6.4
12	Record gaps stay below $\log^2 p$ for $p > 7$	Computation (85 records)	6.6
13	Jet and outer-product identities for ξ	Proposition; checked	7.1, 7.3
14	RH $\iff$ horizontal monotonicity of $ \xi $	Theorem [38]	7.2
15	The functional equation does not force zeros onto the line	Theorem [16]; computed	8.3
16	Spacings match GUE	Conjecture; computed	§9
17	$\sum_{\rho} 1/(\rho(1-\rho)) = 2 + \gamma - \log 4\pi$	Proposition (classical)	9.3
18	The Taylor jet at $\frac{1}{2}$ recovers the zeros	Computation	9.5
19	Ring count falls by 2 per off-line pair	Riemann–von Mangoldt; computed	§10
20	An off-line zero’s prime wave outgrows $\sqrt{x}$	Theorem [41]; modelled	10.3, 10.4
21	Number variance of the zeros saturates	Computation; Berry heuristic	11.1
22	More than two thirds of zeros are simple and on the line	Theorem [1]	§11
23	Q_a sees only $\log n \leq 2a$; λ is nonincreasing	Proposition	12.1, 12.2
24	Matrix floor equals the zero sum	Explicit formula; computed to 2×10^{-5}	12.5
25	Minimizer tracks zeta zeros below $T^*(a)$; zeros above pay λ	Computation; heuristic	12.4, 12.5
26	$-\log \lambda \approx 1.83 T^*(a) + \text{const}$ on $0.8 \leq a \leq 1.1$	Computation (fit)	12.5
27	Zero terms stretch uniformly in x and have fixed frequency in $\log x$	Proposition	13.1
28	Log-mean of $((\psi-x)/\sqrt{x})^2$ tends to $2 + \gamma - \log 4\pi$ under RH	Theorem [13]; 0.0458 at 10^{10}	13.2, 13.3
29	Modular-surface resonances share one decay rate iff RH	Theorem [17, 22]	§13
30	RH	Open	

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